Review

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Traumatic brain injury: Bridging pathophysiological insights and precision treatment strategies

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Abstract

Blood–brain barrier disruption and the neuroinflammatory response are significant pathological features that critically influence disease progression and treatment outcomes. This review systematically analyzes the current understanding of the bidirectional relationship between blood–brain barrier disruption and neuroinflammation in traumatic brain injury, along with emerging combination therapeutic strategies. Literature review indicates that blood–brain barrier disruption and neuroinflammatory responses are key pathological features following traumatic brain injury. In the acute phase after traumatic brain injury, the pathological characteristics include primary blood–brain barrier disruption and the activation of inflammatory cascades. In the subacute phase, the pathological features are characterized by repair mechanisms and inflammatory modulation. In the chronic phase, the pathological features show persistent low-grade inflammation and incomplete recovery of the blood–brain barrier. Various physiological changes, such as structural alterations of the blood–brain barrier, inflammatory cascades, and extracellular matrix remodeling, interact with each other and are influenced by genetic, age, sex, and environmental factors. The dynamic balance between blood–brain barrier permeability and neuroinflammation is regulated by hormones, particularly sex hormones and stress-related hormones. Additionally, the role of gastrointestinal hormones is receiving increasing attention. Current treatment strategies for traumatic brain injury include various methods such as conventional drug combinations, multimodality neuromonitoring, hyperbaric oxygen therapy, and non-invasive brain stimulation. Artificial intelligence also shows potential in treatment decision-making and personalized therapy. Emerging sequential combination strategies and precision medicine approaches can help improve treatment outcomes; however, challenges remain, such as inadequate research on the mechanisms of the chronic phase traumatic brain injury and difficulties with technology integration. Future research on traumatic brain injury should focus on personalized treatment strategies, the standardization of techniques, cost-effectiveness evaluations, and addressing the needs of patients with comorbidities. A multidisciplinary approach should be used to enhance treatment and improve patient outcomes.

Key Words: artificial intelligence; biomarkers; blood–brain barrier; combination therapy; drug delivery; exosomes; focused ultrasound; hyperbaric oxygen therapy; inflammation; nanocarriers; neurodegeneration; personalized medicine; stem cells; therapeutic hypothermia; traumatic brain injury

Introduction

Traumatic brain injury (TBI), a major global public health concern, affects millions of people annually, imposing substantial socioeconomic burdens while severely affecting the quality of life of patients (Maas et al., 2022; Hoh et al., 2024; Xu et al., 2025). TBI can be caused by various mechanical forces, including blunt trauma, penetrating injury, acceleration-deceleration force, and blast injury (Cieri and Ramos, 2025; Seplovich et al., 2025). TBI severity is typically assessed using the Glasgow Coma Scale (McCrea et al., 2021). Based on injury severity, TBI can be classified as mild, moderate, or severe. Mild TBI (commonly known as concussion) is the most prevalent. Moderate TBI typically involves loss of consciousness for 0.5–24 hours and post-traumatic amnesia lasting 1–7 days. Severe TBI is characterized by loss of consciousness exceeding 24 hours and post-traumatic amnesia lasting longer than 7 days (Verboon et al., 2021; Arora et al., 2023). Beyond severity, TBI can also be classified based on injury mechanism (closed or penetrating) and scope (focal or diffuse) (Block et al., 2005; Zingale et al., 2021; Bischof and Cross, 2023). Regardless of severity, TBI induces pathological and functional changes in the brain, with blood–brain barrier (BBB) disruption and a pronounced neuroinflammatory response (Jin et al., 2024).

The BBB is a highly selective physiological barrier that maintains homeostasis in the central nervous system through the regulation of the exchange between circulating blood from brain extracellular fluid. It prevents potentially harmful substances from entering the brain while allowing selective transport of essential nutrients (Profaci et al., 2020; Knox et al., 2022). BBB dysfunction is a hallmark of various neurological disorders (Zhao et al., 2021), including TBI, Alzheimer's disease, epilepsy, and motor neuron disease (Latif and Kang, 2022; Wang et al., 2022; Kiani, 2023). The BBB maintains neural tissue homeostasis through intricate interactions among endothelial cells, astrocytic end feet, and pericytes (Armulik et al., 2010; Sanmarco et al., 2021; Zhang et al., 2021).

Patients with TBI often exhibit long-term cortical BBB dysfunction and persistent neuroinflammation (van Vliet et al., 2020). Neuroinflammation, an inflammatory response in the central nervous system under pathological conditions, involves the activation of brain-resident immune cells and the infiltration of peripheral immune cells (Leng and Edison, 2021; Brandl and Reindl, 2023). The neuroinflammatory response is temporally and spatially regulated, exerting protective and detrimental effects. Initially, it facilitates pathogen clearance and tissue repair in the brain; however, excessive or chronic inflammation inhibits tissue regeneration, leading to neuronal dysfunction and cell death (Cappelletti et al., 2023). Microglia, the primary innate immune cells of

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the central nervous system, plays a central role in neuroinflammation through multiple mechanisms, including the release of pro/anti-inflammatory factors, phagocytosis, and antigen presentation (Peruzzotti-Jametti et al., 2024). They can exert neuroprotective effects by phagocytosing and clearing pathological protein aggregates but potentially produce harmful effects owing to excessive uptake of these aggregates. Therefore, promoting microglial transformation into a protective phenotype is considered an important strategy for treating neurodegenerative diseases (Gao et al., 2023). Astrocytes maintain neuronal function by providing structural and metabolic support while participating in the regulation of neuroinflammatory processes (Lee et al., 2023).

After TBI, the complex interaction between BBB disruption and neuroinflammation exists (Risbrough et al., 2022). BBB disruption permits the entry of inflammatory mediators and immune cells into the central nervous system, exacerbating neuroinflammation, while neuroinflammation further compromises BBB integrity, forming a vicious cycle (Takata et al., 2021). Understanding this dynamic balance is crucial for developing effective TBI treatment strategies. Given the complexity of TBI pathological processes, single therapy often fails to effectively restore this dynamic balance. Therefore, combination therapy, as an emerging therapeutic strategy, shows the potential to disrupt the vicious cycle between BBB disruption and neuroinflammation by simultaneously targeting multiple pathological processes (Caplan et al., 2021b; Rizk et al., 2021). This study innovatively explores the intricate bidirectional regulatory mechanisms between BBB disruption and neuroinflammation following TBI. While significant progress has been made in understanding the relationship between BBB breakdown and neuroinflammation following TBI, key questions in the field remain unanswered, hindering the development of effective therapeutic strategies. Current treatments often fall short because they primarily focus on individual pathways and do not sufficiently address the complex 'vicious cycle' in which BBB damage and neuroinflammation mutually exacerbate each other. Moreover, our understanding of how neuroinflammation dynamically affects neural regeneration following TBI remains insufficient in depth and detail. Therefore, this review aims to provide novel insights into combination therapeutic strategies, with a critical focus on disrupting the vicious cycle between BBB permeability alterations and neuroinflammatory processes. This study aims to elucidate the complex dynamic balance of neuroinflammation in neural regeneration, offering a theoretical foundation for developing more effective TBI treatment strategies. This study addresses the multifaceted pathological mechanisms underlying neural damage and potential regeneration, presenting a paradigm shift from single-target interventions to a more integrated therapeutic approach. Over the past century, TBI research has evolved significantly (**Figure 1**). This progression indicates both a deeper understanding of its pathophysiology and advancements in treatment strategies. Emerging technologies, such as AI-driven precision medicine and advanced combination therapies, are poised to further transform TBI treatment. Understanding this historical progression could provide crucial context for current therapeutic strategies and future directions in TBI research and treatment.

Search Strategy

The literature retrieval was performed by Yujia Lu and Jie Jin using the Google Scholar database. Primary search terms and Boolean combinations included: ("traumatic brain injury" OR "TBI") AND ("blood–brain barrier" OR "BBB"), ("neuroinflammation" OR "inflammation") AND ("TBI" OR "brain injury"), ("combination therapy" OR "treatment") AND ("TBI" OR "brain injury"), ("pathophysiology" OR "mechanism") AND ("TBI" OR "brain injury"), ("imaging" OR "biomarker") AND ("TBI" OR "brain injury"). Additional keywords included: "neuroprotection," "artificial intelligence," "blood–brain barrier," "combination therapy," "stem cells," "drug combinations," "non-invasive brain stimulation," "exosomes," "focused ultrasound," "hyperbaric oxygen therapy," "synthetic nanocarriers," "bone marrow," "extracellular matrix," "energy metabolism," "transmembrane transport," "individualized therapy," "precision medicine," "comorbidities," "monitoring," "gender differences," "age factors," "immune system," "neurodegenerative diseases," "chronic phase," and "rehabilitation." The search was limited to articles published between 2000 and 2024, including research papers, review articles, and clinical trials. Considering the relevance of some older papers, or the limited number of papers on certain topics covered in this review, the latest papers were filtered by year when possible. The search results were further screened based on their relevance to the topic. After excluding unrelated articles and duplicates, approximately 350 articles were selected for this review based on their scientific quality and relevance to the subject matter.

Widespread Physiological Changes After Traumatic Brain Injury

Complex structure of the blood–brain barrier

The existence of the BBB was first demonstrated by Edwin E. Goldmann in 1914 through trypan blue dye experiments (Bentivoglio and Kristensson, 2014). This discovery revealed that the central nervous system possesses selective barrier functions, establishing a crucial foundation for future neuroscience research. The BBB is a dynamic, semi-permeable system that maintains the neural microenvironment homeostasis through precise, selective barrier functions while mitigating therapeutic drug delivery to the central nervous system (Wu et al., 2023a). As an actively regulated biological barrier, the BBB is essential for controlling immune responses within the central nervous system (Qiu et al., 2021). The known mechanisms by which drug molecules cross the BBB include the following: passive transport through paracellular and transcellular diffusion; receptor-mediated transcytosis (e.g., transferrin and insulin receptor-mediated transport); carrier protein-mediated transport (e.g., glucose and amino acid transporters); and adsorption-mediated transcytosis driven by charge interactions (Pandit et al., 2020; Terstappen et al., 2021).

The structure of the BBB is organized from the inside out (**Figure 2**) and consists of the following components: the glycocalyx, endothelial cells, basement membrane (containing pericytes), and astrocytic end-feet (Galea, 2021). The glycocalyx is the innermost layer of the BBB, composed of glycosaminoglycans, including heparan sulfate, chondroitin sulfate, and hyaluronic acid, which cover the luminal surface of blood vessels. It interacts with endothelial cells through membrane-associated CD44 and the proteoglycans syndecan and glypican. This layer is crucial for maintaining BBB integrity, primarily by restricting transcellular transport across brain endothelial cells (Zhu et al., 2022). The glycocalyx directly interacts with blood components while also regulating vascular permeability, facilitating cell signaling, and protecting endothelial cells from mechanical stress (Yang et al., 2021). Compared to the glycocalyx in the peripheral vasculature, the BBB glycocalyx has a distinct glycan composition, physical structure, and surface charge (Dancy et al., 2024). Research has found that systemic inflammation can damage the brain endothelial glycocalyx, leading to significant reductions in thickness and coverage (Yoon et al., 2017).

The endothelial layer of the BBB consists of specialized brain microvascular endothelial cells, which are tightly connected by tight and adherens junctions. These cells are enclosed by a basement membrane primarily composed of type IV collagen, which also surrounds pericytes (Lau et al., 2024). Pericytes are contractile vascular wall cells distributed in the capillaries throughout the systemic microcirculation. In the BBB, pericytes are distributed at an exceptionally high density, a pattern closely related to endothelial barrier function (Li and Fan, 2023). Experimental studies show that acute pericyte loss disrupts normal hemodynamic responses, while extensive pericyte depletion can lead to brain tissue hypoxia and metabolic stress, ultimately altering neuronal excitability and contributing to neurodegeneration (Lau et al., 2024).

In the outermost layer of the BBB, astrocytes secrete a specialized basement membrane through their end feet (Wu et al., 2024). Pericytes and astrocytes function in a complementarity manner within the BBB. Pericytes contribute to BBB integrity by expressing and organizing various tight junction proteins, regulating key transport protein receptors, and facilitating the formation of the capillary basement membrane (Hajal et al., 2021). These components do not function independently; instead, they form a cohesive neurovascular unit along with capillary-associated microglia, perivascular macrophages, and adjacent neurons (Schiera et al., 2024). This intricate structural organization ensures the selective permeability of the BBB and the maintenance of central nervous system homeostasis.

Blood–brain barrier damage triggers inflammatory cascade reactions

Inflammatory response in the central nervous system

Following BBB damage, a sequence of inflammatory cascade reactions unfolds in

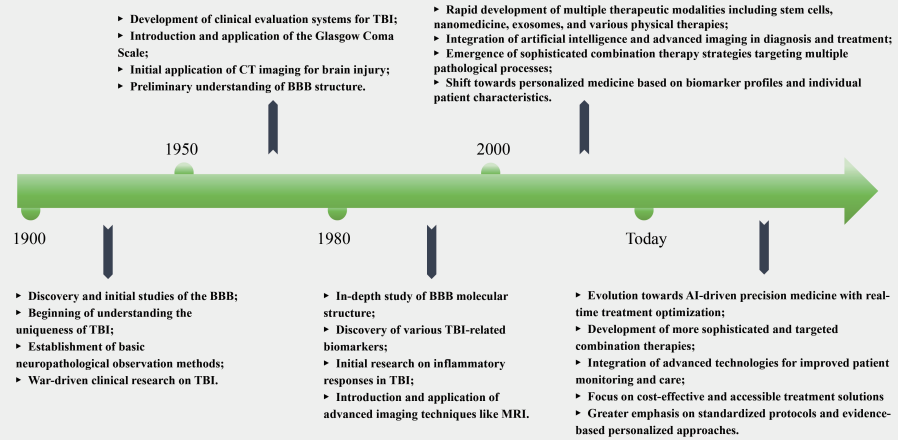

Figure 1 | Historical timeline of key developments in TBI and BBB research.
The timeline illustrates the major milestones in TBI and BBB research from 1900 to the present, along with future perspectives. The progression is divided into four distinct periods: (1) 1900–1950: A foundational period characterized by basic discoveries and war-driven research; (2) 1950–1980: The establishment of clinical standards and diagnostic tools; (3) 1980–2000: The molecular era, marked by an advanced understanding of BBB structure and TBI biomarkers; (4) 2000–present: The modern therapeutic era, featuring multiple treatment modalities and precision medicine approaches. The timeline concludes with future perspectives that emphasize AI-driven precision medicine and advanced therapeutic strategies. A green arrow indicates the chronological progression, while vertical bars highlight significant transition points in the field. AI: Artificial intelligence; BBB: blood–brain barrier; TBI: traumatic brain injury.

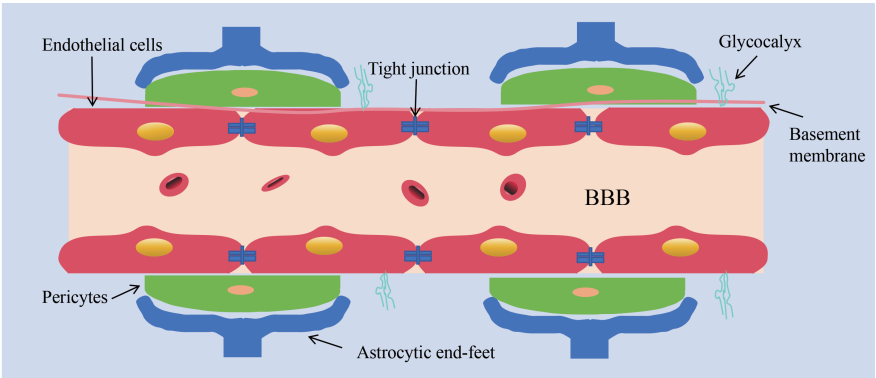

Figure 2 | Structural composition and organization of the BBB.
The illustration depicts the intricate architecture of the BBB, indicating its multi-layered defense system. The barrier consists of several key components arranged in a specific spatial organization: the innermost glycocalyx layer serves as the first line of defense, followed by specialized endothelial cells connected by tight junctions, a surrounding basement membrane, embedded pericytes, and astrocytic end-feet forming the outermost layer. These components work synergistically to maintain homeostasis in the central nervous system through selective permeability control. The tight junctions between endothelial cells are particularly crucial, creating a highly restrictive paracellular barrier that regulates molecular transport. Pericytes and astrocytic end-feet provide both structural support and functional regulation, contributing to BBB integrity and the dynamic response to physiological and pathological conditions. BBB: Blood–brain barrier.

a temporally regulated manner. In the early phase of injury (0–24 hours), central nervous system resident cells, particularly microglia and astrocytes, are the first to be activated (Tabaa et al., 2022). Microglia transition from a resting to an activated state, undergoing significant morphological changes and releasing a wide range of pro-inflammatory cytokines, including tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β), and interleukin-6 (IL-6) (Kong et al., 2024). These cytokines further stimulate microglia and astrocytes through autocrine and paracrine signaling, creating a positive feedback loop that amplifies inflammation. Concurrently, activated astrocytes produce chemokines (such as CCL2 and CXCL1) and adhesion molecules, facilitating the recruitment and infiltration of peripheral immune cells. This process further compromises the already damaged BBB and intensifies the local inflammatory response (Yu et al., 2024).

As the inflammatory response progresses, cytokines and chemokines diffuse beyond the initial injury site, spreading inflammation to surrounding tissues (Cash and Theus, 2020). Pro-inflammatory mediators, especially IL-1β and TNF-α, stimulate damaged BBB endothelial cells to increase the expression of adhesion molecules, including intercellular adhesion molecule-1, vascular cell adhesion molecule-1, and endothelial selectin (Kwon et al., 2023). This upregulation promotes the recruitment of peripheral blood leukocytes (including monocytes and lymphocytes) into the central nervous system through the classical cascade process of contact-rolling-adhesion-transmigration. Once these immune cells infiltrate the brain parenchyma, they release a new round of inflammatory mediators, such as cytokines, reactive oxygen species, and matrix metalloproteinases, further exacerbating the neuroinflammatory response (Kadry et al., 2020).

Participation of the peripheral immune system

Under normal physiological conditions, the brain functions as an immune-privileged organ, remaining largely isolated from the peripheral immune system owing to the presence of an intact BBB (Cui et al., 2022). However, recent studies report that, even under normal conditions, the brain is not fully isolated from the peripheral immune system. A small number of immune cells can cross the BBB through specific mechanisms for immune surveillance (Marchetti and Engelhardt, 2020; Proulx and Engelhardt, 2022). This limited communication plays an important role in maintaining central nervous system homeostasis, but the balance may be disrupted when the BBB is impaired. BBB damage exposes central nervous system–specific antigens to the peripheral immune system. These antigens can be presented by damaged endothelial cells or professional antigen-presenting cells, thereby triggering adaptive immune responses (Mapunda et al., 2022). Moreover, BBB damage permits the massive infiltration of peripheral immune cells, such as monocytes and neutrophils, into the central nervous system. These infiltrating cells not only contribute directly to tissue damage but also exacerbate BBB dysfunction by secreting reactive oxygen species and matrix metalloproteinases, creating a vicious cycle (Fang et al., 2024). Once inside the central nervous system, these immune cells can alter the function of resident glial cells, including microglia, astrocytes, and oligodendrocytes, indicating an intricate network of immune-glial interactions (Croese et al., 2021). This process primarily unfolds during the subacute phase of injury and it is characterized by the activation and clonal expansion of T cells that specifically recognize central nervous system antigens. These activated T cells can breach the damaged BBB and produce additional pro-inflammatory cytokines and cytotoxic factors in the brain, driving the inflammatory response toward chronicity and aggravating neurological dysfunction (Yang et al., 2020). Additionally, astrocyte activation following injury intensifies local inflammation. Astrocytes not only release large amounts of proinflammatory factors but also disrupt neuronal function by altering neurotransmitter metabolism, further amplifying abnormal neural signaling. Furthermore, BBB disruption increases the penetration of complement system components, which enhance innate and adaptive immune responses in the central nervous system (van Erp et al., 2023). Aberrant activation of the complement system induces the release of neurotoxic molecules, such as IL-1β and TNF-α, which directly harm neurons and disrupt distal neural circuits (Ayyubova and Fazal, 2024). Complement activation products, particularly C3a and C5a, promote microglial polarization toward a pro-inflammatory phenotype, thereby exacerbating neuronal damage (Alawieh et al., 2021; Alsbrook et al., 2023). Furthermore, complement deposition destabilizes synaptic structures and impairs neuroplasticity and repair mechanisms, worsening cognitive and motor dysfunction.

The inflammatory cascade reaction triggered by BBB damage is a complex process with temporal and spatial dynamic evolution involving multiple components of the innate and adaptive immune

systems. Resident glial cells within the central nervous system, infiltrating peripheral immune cells, and activated inflammatory signaling pathways interact to form a self-sustaining vicious cycle of inflammation and BBB dysfunction. This complex inflammatory response not only aggravates the initial injury but also contributes to secondary neurological impairment. Consequently, targeting this process represents an important avenue for developing new therapeutic strategies.

Changes in the bone marrow

Bone marrow represents a unique niche intricately connected with the neuroimmune system, exhibiting time-dependent dynamic changes following TBI. These changes involve the regulation of hematopoietic stem cell fate, immune cell generation and migration, cytokine network remodeling, and complex neuroimmune interactions (Qi et al., 2024). The response of bone marrow to TBI reflects a highly coordinated systemic process representing the comprehensive stress response of the body to trauma.

During the acute phase of TBI, bone marrow rapidly responds to trauma signals by initiating adaptive changes. Circulating immune cells—including neutrophils, monocytes, and lymphocytes—migrate extensively to perivascular spaces and the brain parenchyma. While this process aims to facilitate debris clearance and tissue repair, it often worsens neural damage (Goodman et al., 2024). The mobilization and migration of these immune cells are tightly regulated by multiple chemokines, with the CCL2/CCR2 and CXCL12/CXCR4 signaling axes playing crucial roles (Wojcieszak et al., 2022).

Bone marrow hematopoietic stem cells demonstrate significant fate alterations, preferentially differentiating toward myeloid lineages. In the early post-injury phase, hematopoietic stem and progenitor cells rapidly proliferate, leading to increased production of Ly6C^{low} monocytes, which exhibit anti-inflammatory properties. Once these monocytes infiltrate brain tissue, they differentiate into alternatively activated (M2) macrophages, participating in inflammation regulation and tissue repair through IL-10 and TGF-β secretion (Shi et al., 2021)

This dynamic regulation is driven mainly by the sympathetic-adrenal medullary axis and adrenergic signaling pathways. After TBI, sympathetic activation increases norepinephrine release, which activates β-adrenergic receptors in the bone marrow microenvironment. This β-adrenergic signaling stimulates hematopoietic stem cell proliferation and drives anti-inflammatory myeloid cell production for 14 days post-injury. Knockout studies have identified β2 and β3 adrenergic receptors as key regulators of anti-inflammatory macrophage expansion (Mohammadpour et al., 2019; Liu et al., 2023a). This finding provides valuable insights into the regulation of the immune function in the bone marrow.

However, the bone marrow response represents a double-edged sword with time-dependent effects. In the early stages, infiltrating immune cells can intensify neuroinflammation by releasing pro-inflammatory cytokines and ROS. In chronic phases, these cells may accelerate neuronal aging through sustained inflammatory signaling and epigenetic modifications (Ritzel et al., 2018).

In chronic stages, bone marrow function declines, marked by exhaustion of hematopoietic stem cell reserves and reduced proliferative capacity, significantly decreasing leukocyte production (Wu et al., 2023b). This suppression, driven by persistent inflammatory stress and neuroendocrine dysregulation, heightens susceptibility to infections (Ritzel et al., 2023). Simultaneously, alterations in bone marrow stromal cells disrupt HSC maintenance and differentiation.

Recent research has uncovered key insights into the distinct role of calvarial bone marrow in post-TBI immune responses (Li et al., 2024). Immune cells from the calvarial bone marrow communicate directly with the central nervous system through specialized vascular channels and exhibit unique phenotypic and functional characteristics (Soliman et al., 2024). This discovery not only reveals a new source of immune cells but also provides new insights for understanding regional immune regulation in the central nervous system.

The bone marrow microenvironment demonstrates a dynamic transition from protective to pathological effects, involving multiple interconnected systems: the circulatory system mediating immune cell transport, inflammatory pathways regulating cytokine networks, and the lymphatic system participating in immune surveillance (Borlongan and Rosi, 2022). During the acute phase, the bone marrow enhances immune defense through a pro-inflammatory response and mobilizes various immune cells to participate in the injury response. In contrast, during the chronic phase, the bone marrow gradually adopts an anti-inflammatory phenotype, supporting neuroprotection and maintaining inflammatory balance.

Extracellular matrix remodeling

Extracellular matrix (ECM) remodeling represents a core component of BBB dynamic changes following TBI, directly influencing the structural integrity and function of the neurovascular unit (Reed et al., 2019). Post-TBI, ECM components undergo significant spatiotemporal dynamic changes. During the acute phase, matrix metalloproteinases (MMPs), particularly MMP-9 and MMP-2, become highly active, accelerating the degradation of major basement membrane components, including type IV collagen, laminin, and fibronectin (Klein and Bischoff, 2011). This degradation compromises vascular basement membrane integrity, increasing BBB permeability.

The MMP system regulation demonstrates distinct time-dependent characteristics. In the early injury stages, activated glial cells and infiltrating leukocytes rapidly upregulate MMP-9, a key contributor to acute BBB disruption (Vafadari et al., 2016). Subsequently, MMP-2 expression gradually increases, peaking between 3 and 7 days post-injury (Mao et al., 2016). MMP activity is tightly controlled by endogenous inhibitors, and disruptions in this balance directly affect ECM remodeling (Cabral-Pacheco et al., 2020). Therefore, selective MMP inhibitors may provide protection within specific time windows, though complete inhibition could hinder later tissue repair.

ECM remodeling influences BBB integrity in a bidirectional manner. During the acute phase, ECM degradation disrupts tight and adheren junction anchoring, directly compromising endothelial barrier function (Wendt and Gonzales, 2023). Degraded ECM fragments act as damage-associated molecular patterns, triggering inflammatory responses in a positive feedback loop. However, in the subacute and chronic phases, ECM remodeling is essential for tissue repair. Newly synthesized ECM proteins serve as scaffolds for vascular reconstruction and glial scar formation, contributing to neurovascular unit reorganization (Trivedi et al., 2019). The interplay between ECM and neuroinflammation constitutes a complex regulatory network (Ghorbani and Yong, 2021). Degraded ECM components activate inflammatory cells by engaging pattern recognition receptors, such as Toll-like receptors (TLRs), thereby promoting inflammatory factor release. Simultaneously, cytokines secreted by inflammatory cells further upregulate MMP expression, reinforcing a vicious cycle. Certain ECM components, such as specific-sized hyaluronan degradation products, may exert anti-inflammatory effects, highlighting the complexity of ECM remodeling (Marozzi et al., 2021). Post-injury, the ECM affects both the physical barrier function of the BBB and the survival and function of endothelial cells and pericytes by regulating integrin signaling (Aman and Margadant, 2023). Additionally, ECM remodeling regulates the storage and release of neurotrophic factors, a mechanism that may be crucial for long-term neural repair.

Energy metabolic alterations in the blood–brain barrier

Following TBI, the BBB undergoes significant metabolic alterations that compromise both its structural integrity and function (Verweij et al., 2007). As an energy-dependent selective barrier system, the BBB requires a substantial ATP supply to maintain its normal function. Post-TBI metabolic dysregulation directly impairs its ability to maintain homeostasis (Hu and Tao, 2021).

The metabolic alterations in BBB endothelial cells exhibit distinct spatiotemporal characteristics. In the early phase of injury, mechanical influence and cellular stress response induce acute mitochondrial dysfunction, sharply reducing ATP production (Prajapat et al., 2024). ATP depletion disrupts multiple energy-dependent processes, including active transport systems, ion pump function, and cytoskeletal maintenance. This energy insufficiency triggers a cascade of events: first, reduced Na^+/K^+-ATPase activity leading to ionic gradient disruption, causing cellular edema; second, impaired ATP-dependent efflux pumps, such as P-glycoprotein, compromise the capacity of BBB to clear neurotoxic substances; third, insufficient ATP supply hinders the synthesis and proper assembly of tight junction proteins, further weakening BBB barrier integrity (Xu et al., 2021).

During the subacute phase post-injury, metabolic dysfunction perpetuates a vicious cycle. Mitochondrial dysfunction not only reduces ATP production but also increases ROS generation

(Schmitt et al., 2023). Excessive ROS further damages mitochondrial DNA and respiratory chain complexes, exacerbating the energy crisis. Simultaneously, oxidative stress activates endothelial cell death pathways, leading to localized BBB structural destruction. In an attempt to counteract the energy deficit, endothelial cells upregulate glycolysis, but this compensatory shift remains inefficient and may exacerbate local acidosis (Hinca et al., 2021). In the chronic phase, BBB energy metabolism exhibits adaptive changes. Endothelial cells undergo metabolic reprogramming, characterized by reduced mitochondrial numbers but enhanced function, sustained glycolysis activation, and increased fatty acid oxidation. However, these adaptations are often accompanied by decreased metabolic efficiency and reduced energy reserves, which lower BBB tolerance to subsequent stressors (Gribnau et al., 2024). Additionally, metabolic alterations impair BBB active transport, reducing the efficiency of essential nutrient transport, including glucose and amino acids, which subsequently affects neuronal metabolism and function (Lai et al., 2022a).

BBB metabolic alterations involve functional coordination with astrocytes and pericytes (Benaroya, 2020). Under normal conditions, astrocytes support endothelial cells through lactate secretion, but this nutritional support network is disrupted after TBI. Pericytes, as contractile regulations, rely on the local energy supply. When ATP levels decrease, pericyte contractile dysfunction disrupts microvascular tone regulation. Furthermore, complex interactions exist between BBB metabolic alterations and inflammatory responses. Energy depletion activates inflammatory signaling pathways, while inflammation impairs mitochondrial function and energy metabolism, creating a vicious cycle (Amo-Aparicio et al., 2024). This metabolism-inflammation dysregulation worsens acute BBB damage and may also contribute to chronic neurodegeneration.

Transmembrane transport remodeling of the blood–brain barrier

Following TBI, the BBB undergoes extensive remodeling of its transmembrane transport system. This process involves altered transporter protein expression, functional modifications in transport mechanisms, and reorganization of molecular regulatory networks. As a highly selective barrier, the BBB regulates transport through four distinct molecular mechanisms: receptor-mediated transport, carrier-mediated transport, ATP-dependent active transport, and paracellular transport pathways (Lin et al., 2024).

The receptor-mediated transcytosis system undergoes significant restructuring, primarily involving three key receptor complexes. Reduced surface expression of the transferrin receptor and changes in its internalization kinetics disrupt iron homeostasis, potentially worsening neuronal injury through iron-dependent oxidative stress and mitochondrial dysfunction (Stocki et al., 2021). Insulin receptor signaling is impaired, altering insulin-like growth factor transport and compromising endothelial cell survival and function through PI3K/Akt pathway dysregulation (Nagano et al., 2022). The dysfunction of low-density lipoprotein receptor-related protein 1 disrupts bidirectional amyloid-β transport, which may accelerate neurodegeneration (Zhang et al., 2022b).

The remodeling of carrier-mediated transport is highly selective. SLC2A1 (GLUT1) expression and membrane localization undergo dynamic changes, with cytoskeletal reorganization governing membrane trafficking. These alterations disrupt glucose transport, directly influencing energy metabolism (Veys et al., 2020). Dysfunction of the large neutral amino acid transporter LAT-1 (SLC7A5) impairs the supply of essential amino acids, disrupting protein synthesis and reducing neurotransmitter precursor availability. Differential regulation of MCT1 (SLC16A1) and MCT4 (SLC16A3) expression indicates a reorganization of the lactate shuttle system, which plays a critical role in neuron-glial metabolic coupling (Hawly et al., 2024).

The ATP-binding cassette transporter family undergoes complex regulatory changes during BBB remodeling. ABCB1 (P-glycoprotein) dysfunction not only reduces the efflux of neurotoxic substances but also disrupts membrane fluidity by altering the distribution of membrane lipid components (Mohi-Ud-Din et al., 2022). Decreased ABCG2 (BCRP) activity shows a significant correlation with oxidative stress product accumulation. The altered expression of MRP (ABCC) family members affects the transmembrane transport kinetics of glutathione conjugates and inflammatory mediators (Hagos et al., 2019).

The reconstruction of ionic homeostasis involves multiple molecular mechanisms. A selective reduction in the Na^+/K^+-ATPase α2 subunit alters the function of sodium-dependent transporters, further disrupting voltage-dependent transport by affecting membrane potential (Fernandez et al., 2023). Changes in NCX (sodium-calcium exchanger) activity form a vicious cycle with calcium homeostasis disruption while affecting cell adhesion molecule function. Alterations in potassium channel expression, particularly Kir2.1 and Kv1.3, affect ionic balance and contribute to cell volume regulation (Fomina et al., 2021).

The molecular regulatory network governing transport system remodeling is highly complex. At the transcriptional level, transcription factors such as NF-κB and HIF-1α modulate transporter expression by binding to specific promoter elements. Post-transcriptional regulation involves miRNA-mediated control of mRNA stability and inhibition of protein translation. At the protein level, regulation occurs through selective degradation via the ubiquitin-proteasome system and the dynamic balance between endocytosis and recycling (Mi et al., 2021). These remodeling processes have significant implications for therapeutic drug delivery. Traditional BBB penetration strategies require reevaluation and optimization in response to dynamic transport changes. Additionally, impaired endogenous clearance pathways may cause neurotoxic substances to accumulate, triggering secondary injury cascades.

Dynamic Balance Between Blood–Brain Barrier Permeability and Neuroinflammation

TBI can cause various neurological and neuropsychiatric complications, including cognitive decline, depression, anxiety, and epilepsy, alongside symptoms such as headache and sleep disorders (Wangler and Godbout, 2023). Based on pathophysiological characteristics, the injury process after TBI can be divided into three stages: the acute (within hours after trauma) (**Figure 3**), subacute (days to weeks), and chronic phases (weeks and beyond) (Balança et al., 2021; Hoffe and Holahan, 2022; Amlerova et al., 2024). The dynamic balance between BBB permeability and neuroinflammation after TBI is crucial in determining patient prognosis (Padmakumar et al., 2022). Disruption of this balance can lead to different pathological outcomes: persistent BBB disruption facilitates continuous influx of inflammatory mediators into the central nervous system, while excessive neuroinflammation inhibits repair mechanisms, compromising neuronal survival and functional recovery.

Acute phase

Within 24 hours of TBI occurrence, primary injury manifests as hemorrhage, shock, and arterial hypotension resulting from mechanical forces (Picetti et al., 2019). BBB disruption initially occurs in the impact core region, allowing blood components and cells to infiltrate the brain parenchyma, triggering cerebral edema and neuroinflammatory responses (Koyama, 2021). This disruption permits the leakage of potentially harmful substances, such as serum proteins, into the brain parenchyma, exacerbating injury severity. The acute phase inflammatory response is driven by neuroglial cell activation, with microglia and astrocytes playing key roles in detecting and responding to injury (Tang and Harte, 2021). After trauma, microglia rapidly proliferate and migrate to the injury site, releasing large amounts of inflammatory mediators and amplifying the inflammatory cascade (Willing et al., 2020). A positive feedback loop forms between BBB disruption and inflammatory response—inflammatory cytokines promote further BBB breakdown, increasing immune cell infiltration and perpetuating the injury-inflammation cycle (Prabhakar et al., 2022). Additionally, peripheral immune cells and inflammatory factors infiltrate the central nervous system through the damaged BBB, further exacerbating neuroinflammation (Datta et al., 2023). The expression levels of various inflammatory factors, such as TNF-α, IL-1β, and IL-10, among others., fluctuate dynamically with TBI response, and they are identified as potential biomarkers for TBI diagnosis and prognosis (Werhane et al., 2017). Additionally, the ECM in the cortex undergoes changes, with fibronectin levels significantly decreasing at 15 minutes, 1 hours, and 2 hours post-TBI, while hippocampus tenascin-C levels increase significantly at 15 minutes post-injury (Griffiths et al., 2020).

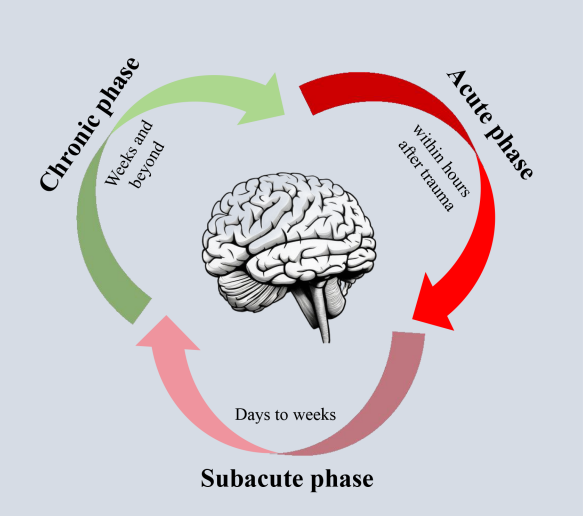

Figure 3 | Temporal progression of TBI through distinct pathophysiological phases.
The illustration depicts the dynamic progression of TBI through three critical phases: acute, subacute, and chronic. The acute phase occurs within hours after trauma and is characterized by primary injury mechanisms and immediate inflammatory responses. The subacute phase spans days to weeks, representing a transition period in which repair mechanisms begin to activate while secondary injury processes continue. The chronic phase extends for weeks and potentially lasts for years, during which long-term pathological changes and ongoing repair processes persist. The circular arrangement of arrows emphasizes the continuous nature of these phases and their interconnected pathophysiological processes, suggesting that intervention strategies should be tailored to each specific phase while considering the overall temporal progression of the injury. The central illustration of the brain highlights that these phases affect the entire organ system, influencing both structural and functional outcomes. TBI: Traumatic brain injury.

Subacute phase

During the subacute phase, the body activates multiple regulatory mechanisms to restore homeostasis. Endothelial cells, the primary structural component of the BBB, play a central role in the repair. Tight junction proteins (claudin-5, occludin, and ZO-1), which degrade during the acute phase, gradually upregulate and reorganize during the subacute phase, contributing to BBB restoration. Astrocytes promote BBB integrity recovery by secreting anti-inflammatory cytokines and neuroprotective factors (such as TGF-β1) (Rosa et al., 2021). Chondroitin sulfate proteoglycans regulate fibroblast growth factor 2 (FGF2) activity, maintaining neural stem cell undifferentiation. During the subacute phase following TBI, FGF2 expression significantly increases in brain tissue, creating a therapeutic window for enhanced neuroprotective interventions (Betancur et al., 2017). Concurrently, platelet-derived growth factor receptor β (PDGFRβ) expression is upregulated within the first-week post-TBI, accumulating in cortical and thalamic regions near the injury. PDGFR β-positive BBB-related cells can persist in damaged brain regions for months (Kyyriäinen et al., 2017).

During the subacute phase, peripheral immune cells also undergo functional transformation. Infiltrating macrophages gradually transform into a phenotype promoting tissue repair, while regulatory T cells infiltrate the central nervous system, secreting anti-inflammatory factors to resolve inflammation (Wang et al., 2023b; Xu et al., 2023). Microglia gradually transform from the pro-inflammatory M1 phenotype to the anti-inflammatory M2 phenotype, increasing the secretion of anti-inflammatory cytokines (IL-10, IL-4, and TGF-β) (Li et al., 2022c). Additionally, M2 macrophages secrete anti-inflammatory lipid mediators, facilitating the transition of immune responses from pro-inflammatory to resolution states (Yan et al., 2024). However, the dichotomous classification of microglia as expressing only M1 or M2-specific markers is challenging. During TBI pathology, M1/M2 microglial polarization potentially represents a dynamic, coexisting process rather than mutually exclusive states (Lyu et al., 2021). The presence or absence of microglia does not directly affect cognitive deficits in TBI mice. Their neuroprotective effects are highly time-dependent. They exert positive effects only within specific microenvironments during the acute injury phase, potentially losing this protective effect in later acute injury stages (Willis et al., 2020).

Chronic phase

The chronic phase of TBI progresses more gradually than the acute and subacute phases but involves persistent pathological changes (Bashir et al., 2020). Neurofilament light protein and glial fibrillary acidic protein in the blood remain elevated for years post-injury, while tau protein is commonly elevated in chronic TBI patients with neurodegenerative diseases (Friberg et al., 2024). Even after TBI symptom remission, underlying molecular pathological processes persist, often contributing to age-related neurodegenerative diseases (Saikumar and Bonini, 2021).

BBB integrity remains compromised in the chronic phase, failing to fully recover to its pre-injury state (Cherian et al., 2019). Although permeability is significantly lower than in the acute phase, persistent disruption allows continued infiltration of peripheral immune cells and inflammatory molecules, maintaining chronic neuroinflammation (Willing et al., 2020).

At the molecular level, chronic BBB dysfunction is characterized by sustained dysregulation of transport systems and barrier proteins. Altered expression of tight junction proteins claudin-5 and occludin impairs barrier function, while efflux transporter P-glycoprotein dysfunction may hinder the ability of the brain to clear toxic substances (Yamazaki et al., 2019; Zhang et al., 2023). At the immune cell level, chronic phase microglia exhibit dynamic transformation between pro-inflammatory and anti-inflammatory states (Yassaghi et al., 2024). Even low-level persistent pro-inflammatory factor expression can potentially disrupt neuronal function and synaptic plasticity over time, contributing to cognitive and behavioral disorders (Duffy et al., 2018). Astrocytes play a dual role in maintaining chronic phase dynamic balance, and they support BBB repair by promoting new tight junction formation. In contrast, continuously activated astrocytes may transform into a chronic reactive phenotype, sustaining neuroinflammatory states and facilitating the occurrence and development of neurodegenerative changes (Lee et al., 2022; Friberg et al., 2024).

Influential factors

The dynamic balance between BBB permeability and neuroinflammation after TBI is regulated by multiple factors, including genetic factors, age, sex differences, and environmental factors (Anto-Ocrah et al., 2021; Blommer et al., 2021; Antrobus et al., 2022). First, genetic factors play a key role in regulating BBB integrity and inflammatory responses after brain injury (Duchniewicz et al., 2024). Most typical among these is APOE gene polymorphism: individuals carrying the e4 allele face a higher risk of BBB disruption and vascular leakage, making it the strongest known genetic risk factor for Alzheimer's disease development (Atherton et al., 2022; Reddi et al., 2022).

Second, age significantly influences TBI prognosis. Young individuals exhibit faster BBB repair and more efficient inflammation regulation (Izzy et al., 2022). In contrast, aging individuals show significantly weakened BBB repair ability and chronic low-grade inflammation, increasing their susceptibility to cognitive decline and dementia (Bowman et al., 2018; Oroszi et al., 2024).

Sex differences also influence BBB permeability and neuroinflammatory responses. Pre-menopausal females generally show better TBI outcomes, owing to the neuroprotective effects of estrogen, which suppress pro-inflammatory factors and enhance anti-inflammatory signaling, thereby enhancing BBB integrity and repair (Moraga-Amaro et al., 2018; Blaya et al., 2022; Tarudji et al., 2023). Males often exhibit stronger and more prolonged inflammatory responses (Baskin et al., 2023). Post-menopausal women lacking estrogen protection show risk and recovery patterns similar to males, with hormone decline linked to increased chronic low-grade inflammation (Blaya et al., 2022). However, some studies report that women are more susceptible to persistent TBI-related cognitive and somatic symptoms (Levin et al., 2021; Blaya et al., 2022) or minimal difference in dementia diagnosis risk between sexes post-TBI (Kornblith et al., 2020).

Beyond sex hormones, other hormones significantly influence the BBB. Corticotropin-releasing Hormone induces mast cell degranulation under stress, disrupting the BBB (Bhuiyan et al.,

2021). Corticosterone or cortisol, key hormones regulating glucose metabolism and immune responses, exacerbate TBI-induced neuronal injury through multiple mechanisms, including glucose transport inhibition, calcium metabolism alteration, and neurotrophin expression suppression (Komoltsev and Gulyaeva, 2022). Additionally, certain gastrointestinal hormones secreted by intestinal endocrine cells can cross the BBB and diffuse into the central nervous system, though their specific mechanisms remain unclear (Wu et al., 2022).

Environmental factors, including lifestyle and toxin exposure, also significantly influence the BBB-inflammation dynamic balance. High saturated fat and sugar diets increase oxidative stress and inflammation levels, impairing BBB function (Ledreux et al., 2016). Conversely, diets rich in antioxidants and omega-3 fatty acids help maintain BBB integrity and reduce neuroinflammation levels (Pelgrim et al., 2022). Environmental toxin exposure disrupts BBB function by triggering oxidative stress and inflammatory cascade reactions, delaying recovery (Ni et al., 2022).

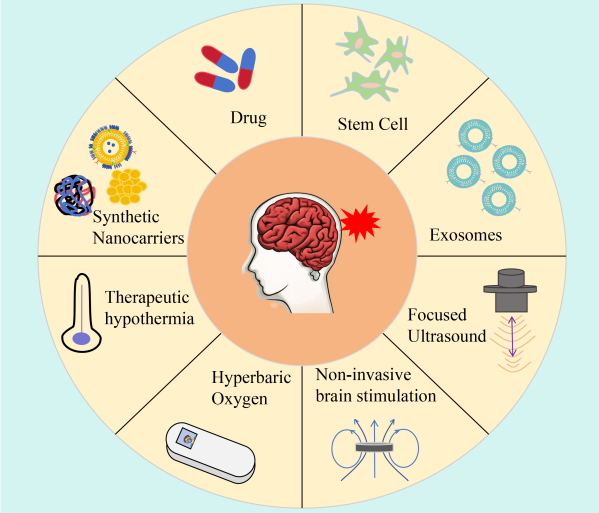

Figure 4 | Comprehensive therapeutic approaches for TBI.
The diagram illustrates the eight principal therapeutic strategies currently used in the treatment of TBI, arranged in a circular pattern around a central image of the affected brain. These interventions include conventional drug therapy, stem cell-based treatments, exosome therapy, focused ultrasound technology, non-invasive brain stimulation, hyperbaric oxygen therapy, therapeutic hypothermia, and synthetic nanocarrier delivery systems. Each modality represents a distinct therapeutic approach targeting different aspects of TBI pathophysiology, ranging from cellular repair and regeneration to blood–brain barrier modulation and regulation of the neuroinflammatory response. The circular arrangement emphasizes the potential for combinatorial therapeutic approaches, where multiple strategies can be integrated to achieve optimal treatment outcomes. This comprehensive therapeutic framework reflects the complex and multifaceted nature of TBI pathology and the need for diverse intervention strategies. TBI: Traumatic brain injury.

Combination Therapy After Traumatic Brain Injury

Although preventive strategies are ideal for reducing primary injury, therapeutic strategies targeting secondary injury directly intervene to reduce the cascade of destructive events (Li et al., 2021b). The complex interactions of various cellular and molecular events after TBI (**Figure 4**) have hindered the development of a universally effective treatment strategy (Lynch et al., 2023). Current TBI treatment options include both pharmacological and non-pharmacological treatments. Pharmacological treatments mainly include neuroprotective agents, antidepressants, anticonvulsants, and symptomatic medications for complications. Non-pharmacological treatments involve rehabilitation therapy, physical therapy, and cell transplantation (Song et al., 2024). Based on the previous in-depth analysis of the bidirectional regulation between the BBB and neuroinflammation, TBI treatment strategies can be explored through several aspects: developing drugs and targeted delivery systems for protecting and repairing the BBB, developing specific drugs targeting neuroinflammation, and designing comprehensive treatment plans addressing both the BBB and neuroinflammation. This synergistic treatment approach emphasizes the importance of multi-target intervention, highlights the possibility of reducing single-drug side effects, and achieves individualized treatment (Malfitano et al., 2020).

Drug combinations

The basic principle of combination drug therapy is to improve therapeutic efficacy and overcome treatment resistance by utilizing drugs with known activity, unique mechanisms of action, and minimal overlapping toxicity characteristics (Löscher and Klein, 2022). This approach is particularly significant in treating complex neurological diseases, where single-target interventions often fail to sufficiently alter underlying pathophysiological processes to significantly affect disease progression (Calzetta et al., 2024). Currently, no single drug intervention can effectively prevent neuronal damage or promote neural recovery after TBI (Akira et al., 2022). However, evidence suggests that multi-target treatment strategies may drive breakthrough progress (**Table 1**). Preclinical studies show that combining minocycline with N-acetylcysteine has positive synergistic effects with extended treatment windows after TBI compared to those of single drugs (Ghiam et al., 2021; Lawless and Bergold, 2022). In a landmark study involving 183 patients with TBI, researchers compared dual therapy (vitamin D3 and progesterone) with quadruple therapy (vitamin D3, omega-3 fatty acids, glutamine, and progesterone). The results showed that quadruple therapy significantly improved neurological function recovery: the recovery rate of Glasgow Coma Scale scores reaching 10 increased to 90% (compared to 60% with dual therapy), mortality decreased to 6.6% (compared to 10% with dual therapy), strongly confirming the synergistic neuroprotective effects of multi-pathway regulation (Matthews et al., 2020).

Furthermore, mechanistic studies show effective treatment combinations that target different pathophysiological processes (Hiskens, 2022; Lynch et al., 2023). For example, acute combination therapy with intravenous memantine and 17β-estradiol demonstrate neuroprotective effects post-TBI through enhanced neuronal survival and attenuated neuronal mortality (Day et al., 2017). The combined administration of simvastatin and recombinant human erythropoietin exhibits significant synergistic effects: recombinant human erythropoietin promotes cell proliferation, while simvastatin promotes axonal repair, leading to superior structural and cognitive function recovery than that of monotherapy (Chauhan and Gatto, 2010). Similarly, combining N-acetylcysteine with probenecid (a membrane transport protein inhibitor) improves the limited BBB permeability and brain tissue retention of NAC (Clark et al., 2023). Regarding neuroinflammation, single anti-inflammatory drugs (such as IL-1ra inhibitors, COX-2 inhibitors, or microglial caspase-1 antagonists) showed limited efficacy. However, combining IL-1ra and COX-2 inhibitors significantly reduced acute-phase CA1 region damage and effectively suppressed epilepsy occurrence (Kwon et al., 2013).

Recent advances in materials science and nanotechnology enable sophisticated BBB permeability regulation and brain-targeted drug delivery systems (Wu et al., 2023a). These technological innovations further enhance the therapeutic potential of combination therapy by improving drug bioavailability and targeting efficiency. The success of multi-target therapies extends beyond TBI. A large-scale veteran study reports that combining metformin-ACEI-β-blockers significantly reduces the risk of Alzheimer's disease (Wang et al., 2023a), providing new insights for treating neurodegenerative diseases.

The Glasgow Coma Scale (GCS) remains the gold standard for assessing neurological function in patients with TBI for nearly 50 years. The scale has undergone multiple optimizations and adjustments based on parameters including age, pupillary response, and brainstem reflexes (Bajaj et al., 2023). GCS is crucial for quantifying TBI severity and evaluating the necessity for surgery. According to the scoring criteria, GCS scores of 13–15 indicate mild TBI, 9–12 moderate TBI, and ≤ 8 severe TBI (Bick et al., 2022). Currently, a GCS score ≤ 10 serves as a significant indicator for level one trauma, as greater TBI severity positively correlates with mortality risk.

Table 1 | Overview of combination drug therapy strategies for TBI

Treatment phases	Combination types	Therapeutic objectives	Representative combinations	Primary benefits	Key considerations
Acute phase	Neuroprotective combinations	Reduce secondary injury and stabilize neuronal function	NMDA receptor antagonists + neurosteroid, calcium channel blockers + free radical scavengers	Reduce neuronal death, controlled intracranial pressure, stabilized blood–brain barrier	Blood pressure monitoring, electrolyte balance, ICP monitoring
Subacute phase	Neuroregenerative combinations	Promote neural repair and improve circulation	Neurotrophic factors + antioxidants, circulatory enhancers + energy substrates	Enhanced neuroregeneration, improved cerebral blood, attenuated inflammation	Hepatorenal function, coagulation, infection surveillance
Recovery phase	Functional Reconstruction Combinations	Facilitate functional recovery and improve prognosis	Cognitive enhancers + neuromodulators, metabolic optimizers + nutritional support	Enhanced cognitive function, improved motor recovery, elevated quality of life	Long-term efficacy, complication prevention, rehabilitation compliance

This table outlines the phase-specific characteristics and clinical applications of combination therapy strategies for TBI. It systematically details various types of drug combinations, therapeutic objectives, representative combinations, primary benefits, and key considerations across three phases: acute, subacute, and recovery. This phased treatment framework provides essential guidance for clinicians in developing individualized treatment protocols, optimizing therapeutic timing, and designing effective drug combinations, ultimately enhancing treatment outcomes for TBI. The comprehensive overview facilitates evidence-based decision-making in clinical practice and supports the development of targeted therapeutic interventions. ICP: Intracranial Pressure; NMDA: N-methyl-D-aspartate; TBI: traumatic brain injury.

Following TBI, increased intracranial air content and blood volume can elevate Intracranial Pressure (ICP) and impair cerebral perfusion, the primary cause of death in most patients with TBI (Wafie et al., 2023). Normal ICP values should be maintained within 5–15 mmHg; intracranial hypertension is defined as ICP persistently exceeding 20 mmHg for over 5 min. ICP monitoring can be performed through ventricular catheterization or parenchymal monitors (Freeman, 2015). Cerebral Perfusion Pressure (CPP), calculated as the difference between mean arterial pressure and ICP, along with ICP abnormalities, is closely associated with BBB damage and neuroinflammation. They significantly affect the neurological status of patients, typically reflected in GCS score changes (Shim et al., 2023; Wang et al., 2024c). In patients with lower GCS scores, implementing combination therapy protocols based on dynamic ICP and CPP monitoring can significantly optimize treatment outcomes. Specific measures include using osmotic diuretics (such as mannitol) to reduce ICP, applying vasoactive drugs (such as nimodipine) to improve CPP, both of which directly alleviate brain tissue compression and ischemic injury, improving neuronal function (Syzdykbayev et al., 2024). Simultaneously, combining IL-6 receptor antagonists and other anti-inflammatory drugs with BBB repair agents such as iron chelators or antioxidants mitigates neuroinflammation and barrier disruption at the pathological level, promoting neurological function recovery. Dynamic tracking of GCS score changes enables timely evaluation of treatment effectiveness and adjustment of treatment plans accordingly, improving patient outcomes.

In addition to these monitoring methods, brain tissue oxygen tension monitoring, jugular venous oxygen saturation measurement, and intracerebral microdialysis neurochemical monitoring (Musick and Alberico, 2021). Given the limitations of single monitoring modalities, the integrated application of multimodal neurological monitoring is essential, as each monitoring method provides unique and clinically relevant insights, complementing each other for a more comprehensive assessment (Citerio et al., 2015).

Hyperbaric oxygen therapy

Hyperbaric oxygen therapy (HBOT) is a treatment method that involves inhaling pure oxygen under elevated atmospheric pressure (1–3 ATA). Research highlights that this method effectively alleviates secondary damage and inflammatory responses after TBI (Jiang et al., 2021). Currently, the FDA approved HBOT for treating specific conditions, including decompression sickness and CO poisoning. Although some medical institutions promote its use for Alzheimer's disease and stroke treatment, its clinical efficacy remains under investigation (Biggs et al., 2021).

HBOT participates in the TBI treatment process through multiple pathways. Clinical research reveals that this therapy reduces intracerebral hematoma volume, promotes normalization of brain electrical activity, and regulates the expression levels of various biomarkers. These indicators include NSE, S100β, and glial fibrillary acidic protein (GFAP), whose expression are closely associated with 'the cognitive function and consciousness recovery of patients (Chen et al., 2022). At the molecular level, HBOT mainly inhibits neuroinflammation by regulating signaling pathways such as p38-MAPK-CCL2 and NF-κB/MAPKs-CXCL1 (Jiang et al., 2021; Xia et al., 2022). HBOT enhances combination therapy by significantly boosting the therapeutic effects of nanopharmaceuticals such as Doxil and Abraxane (Liu et al., 2021). This synergistic effect highlights the potential of HBOT in TBI treatment, suggesting its use in combination with other treatment modalities for improved outcomes.

Focused ultrasound

Focused ultrasound (FUS) technology, particularly microbubble-mediated FUS, is a non-invasive method for temporarily opening the BBB, enhancing therapeutic drug accumulation in target brain regions (Martinez et al., 2024). Despite skull-induced ultrasound attenuation, high-intensity focused ultrasound can precisely target deep brain tissue without damaging adjacent structures. Under low acoustic pressure conditions, FUS utilizes microbubbles to amplify the mechanical effects of sound waves on blood vessels, achieving controlled BBB opening (Pandit et al., 2020). FUS-mediated BBB opening involves two key processes: increasing passive diffusion through loosening tight junctions and promoting active transport through enhanced endocytosis and induced transendothelial fenestration (Beccaria et al., 2020).

To complement FUS-based drug delivery, researchers developed advanced systems. For example, novel perfluorooctyl bromide nanoemulsions increase the brain distribution of hydrophobic drugs after FUS-mediated BBB opening (Bérard et al., 2022). In central nervous system disorders, clinical trials show that focused ultrasound with microbubbles effectively induces targeted immune effects and provides immunotherapy (Jung et al., 2022). Recent research focuses on developing MRI-guided FUS systems, ultrasound-responsive nanocarriers, and optimizing treatment protocols based on individualized parameters. However, even a temporary BBB opening may affect neurovascular unit integrity, posing a risk to healthy brain tissue (Banks et al., 2024). Therefore, current research prioritizes precise control of ultrasound parameters, the development of real-time monitoring methods, and the establishment of standardized operating procedures. The future of ultrasound-based synergistic therapies for brain disorders is promising (Choi and Kim, 2021).

Noninvasive brain stimulation

Noninvasive brain stimulation (NIBS) is increasingly used in rehabilitation settings, demonstrating therapeutic potential for various neurological conditions, including stroke, spinal cord injury, TBI, and multiple sclerosis. The two primary NIBS modalities are Transcranial Magnetic Stimulation (TMS) and transcranial Direct Current Stimulation (tDCS) (Kesikburun, 2022). Both techniques modulate cortical excitability and synaptic plasticity, inducing long-term potentiation and long-term depression-like changes (Markowska and Tarnacka, 2024).

TMS uses electromagnetic induction to generate rapidly changing magnetic fields over the scalp to stimulate cortical neurons (Cappon et al., 2022). TMS protocols are categorized into high-frequency (≥ 5 Hz) and low-frequency (< 1 Hz) stimulation, which induce excitatory and inhibitory effects, respectively. In contrast, tDCS delivers weak direct currents through scalp electrodes, where anodal stimulation typically facilitates, and cathodal stimulation inhibits cortical excitability (Yoon et al., 2016).

Clinical evidence suggests that NIBS can be effectively utilized as a standalone intervention or in combination with other therapeutic approaches to enhance treatment outcomes (Galimberti et al., 2024). In depression treatment, combining antidepressant medications with either rTMS or tDCS demonstrates superior efficacy in alleviating depressive symptoms than monotherapy (Liu et al., 2017; Tao et al., 2024). This synergistic effect

may be attributed to the concurrent modulation of neuroplasticity by both pharmacological and stimulation interventions. Furthermore, combining neuroplasticity-inducing pharmacological agents with NIBS techniques targeting the primary motor cortex shows promise for treating various neurological disorders (Perez et al., 2014).

Therapeutic hypothermia

Hypothermia therapy is an intervention strategy that prevents and treats brain injury by lowering core body temperature. Reducing the temperature several degrees below normal slows blood flow, suppresses inflammation, and reduces tissue metabolic demands, thereby exerting neuroprotective effects (Zhang et al., 2022a). However, the clinical application of hypothermia treatment remains controversial, including the choice between whole-body and local hypothermia, the time window between short-term and long-term hypothermia, and treatment evaluation criteria. Currently, clinical research lacks a unified consensus (Trieu et al., 2023; Lavinio et al., 2024).

Nevertheless, basic research highlights the potential value of hypothermia treatment. In a study using a severe TBI rat model, long-term hypothermia significantly enhanced neuroprotective effects by antagonizing intracranial pressure rebound (Sun et al., 2024). Combining hypothermia with other treatment modalities enhances synergistic effects. For example, sequentially combining mild hypothermia with recombinant high-density lipoprotein effectively improved multiple pathological indicators in TBI mouse models compared to that of hypothermia treatment alone. These include inflammation levels, oxidative stress status, neuronal survival rate, and BBB integrity, leading to significant improvement in spatial learning and memory abilities (Huang et al., 2023). Another study confirms that combining hypothermia with intranasal insulin administration effectively prevents volume increase and cell loss in hippocampal CA1 and DG subregions, thereby improving anxiety-like behavior and memory deficits after TBI (Jahromi et al., 2024).

Stem cell therapy

Mesenchymal stem cells (MSCs), obtained from various tissues, including bone marrow, umbilical cord, adipose tissue, and placenta, are promising therapeutic candidates for TBI (Lin et al., 2023b). Although exogenous stem cell transplantation can regulate inflammatory responses and promote endogenous neurogenesis through the secretion of growth factors and cytokines (Guo et al., 2024), the persistent inflammatory response and neurodegenerative cascade reactions in the TBI microenvironment significantly affect transplanted cell survival, particularly without auxiliary interventions (Willing et al., 2020).

Research reveals that combination therapies can effectively mitigate these limitations. Mahmood et al. report that MSCs combined with simvastatin at optimal dose ratios significantly improved functional outcomes compared to monotherapy, confirming their synergistic effects in TBI treatment (Mahmood et al., 2008). This synergistic effect has been verified across various combination regimens. Exogenous basic fibroblast growth factor significantly accelerates and enhances the therapeutic effects of bone marrow-derived MSC transplantation (Liu et al., 2014). In chronic severe TBI, stem cell factor combined with granulocyte colony-stimulating factor promotes myelin regeneration in the ipsilateral external capsule and striatum (Qiu et al., 2023). Additionally, the combination of β-blocker propranolol with MSCs showed more significant improvement in cognitive memory function than monotherapy (Kota et al., 2016). Additionally, regulatory T cells (Treg) combined with MSCs showed stronger anti-inflammatory effects than therapy alone (Caplan et al., 2021b).

Preclinical studies report the significant efficacy of various innovative combination therapy regimens (Kline et al., 2016; Ahmed, 2022). In the lateral fluid percussion injury model, combining pioglitazone with human MSCs reduced brain injury volume, neurodegeneration, and glial cell proliferation while decreasing inflammatory chemokine expression and increasing brain-derived neurotrophic factor levels and subventricular zone neurogenesis (Das et al., 2019). Another study used transgenic human umbilical cord MSCs overexpressing the CXCR4 receptor, combined with BDNF-modified chitosan scaffolds, effectively enhanced transplanted stem cell migration to injury boundaries (Huang et al., 2016). An innovative study explored the combined application of human neural stem/progenitor cell transplantation with oral curcumin-PLGA nanoparticles, significantly improving motor function, reducing cerebral edema, and suppressing astrocyte proliferation and inflammatory factor expression (Narouiepour et al., 2022). Furthermore, the combined application of shCCL20-CCR6 nanodendrimer complexes with hMSCs more effectively increased brain-derived neurotrophic factor levels and alleviated TBI symptoms than single treatments (Mayilsamy et al., 2020).

Stem cells are crucial for BBB reconstruction. Research reveals that human bone marrow MSCs contribute to tight BBB reconstruction by functioning as perivascular pericytes (Kim et al., 2021). Studies show that stem cells can differentiate into various BBB cellular components (Qian et al., 2017; Nishihara et al., 2020). Human-induced pluripotent stem cells can differentiate into brain microvascular endothelial-like cells with mature immunophenotypes, highlighting their therapeutic potential (Matsuo et al., 2023). However, despite extensive *in vitro* evidence suggesting that combination therapy could enhance therapeutic effects and immunosuppressive capabilities, its effectiveness in improving BBB function remains controversial. For instance, Caplan et al. (2021a) report that combining T-cell and MSC therapy showed no significant advantages over individual treatments in improving BBB permeability or enhancing endogenous immune responses *in vivo*. This phenomenon may stem from factors including cell infusion timing and administration protocols. Therefore, further systematic research is needed to determine optimal combination therapy regimens and their clinical application value.

Exosomes

Exosomes are naturally occurring nano-sized vesicles (30–150 nm) enclosed by lipid bilayers carrying various bioactive molecules, including proteins, lipids, and nucleic acids (Ollen-Bittle et al., 2024). Exosomes can be used independently for TBI treatment and also serve as delivery systems in innovative combination therapies.

Exosomes possess the ability to penetrate the BBB while showing good biocompatibility and low immunogenicity, making them ideal carriers for delivering therapeutic substances to the central nervous system (Zhong et al., 2023). Based on this characteristic, researchers have developed various exosome-drug combination regimens. For example, in a zebrafish brain cancer model, intravenously injected exosomes loaded with anticancer drugs (doxorubicin and paclitaxel) could effectively enter the brain while free drugs fail to cross the BBB (Yang et al., 2015). Therapeutic agents can be combined with exosomes to form combination preparations through multiple approaches, including direct co-incubation with isolated exosomes or transfection of donor cells to secrete exosomes carrying target substances (Li et al., 2022a). For instance, researchers successfully reduced stem cell stress injury and promoted their differentiation into neurons by loading brain-derived neurotrophic factors into exosomes derived from human neural stem cells (Zhu et al., 2023). In another study, systemically administered macrophages transfected with catalase plasmid DNA secrete extracellular vesicles carrying catalase-related genetic material (pDNA, mRNA) and active enzymes, significantly improving neurological function (Haney et al., 2013).

The synergistic application of exosomes with biomaterials offers a promising approach to combination therapy. Research reveals that integrating bone marrow MSC-derived exosomes into hyaluronic acid-collagen hydrogels synergistically promotes neural stem cell differentiation into neurons and oligodendrocytes while inhibiting astrocyte differentiation (Li et al., 2022b; Liu et al., 2023b). Combining human neural stem cell-produced exosomes with three-dimensional nanoscaffolds containing SDF1α bio-sequences significantly reduces oxidative stress levels in serum and brain tissue after TBI (Hajinejad et al., 2023). Currently, researchers have developed various biomaterials and manufacturing technologies to create three-dimensional scaffolds that promote neural injury repair. These scaffolds sustainably release exosomes containing regenerative factors, including neurotrophic factors, mRNA, and miRNA, showing significant therapeutic effects in animal models (Poongodi et al., 2021).

Exosomes exhibit unique therapeutic effects. Depending on their source, they regulate neuroinflammatory responses and reduce neuronal injury through their specific bioactive molecules (Jin et al., 2024; Xiong et al., 2024). For example, human MSC-derived exosomes, enriched with regulatory miRNAs, inhibit pro-inflammatory cytokine release after TBI (Willing et al., 2020). Exosomes secreted by activated astrocytes inhibit the NF-κB signaling pathway through their carried miRNAs, thereby reducing microglia-mediated neuroinflammation (Long et al., 2020). In a porcine

TBI model, early single-dose treatment with human MSC-derived exosomes promoted neuroprotection and enhanced BBB integrity (Williams et al., 2020). These inherent biological effects, combined with their carrier function, establish exosomes as a key platform for developing TBI combination therapy strategies.

Synthetic nanocarriers

The BBB, a natural barrier of the central nervous system, protects the brain environment but also limits therapeutic drug delivery. Many Phase III clinical trials targeting TBI have failed primarily due to low drug targeting efficiency and insufficient brain retention (Mohammed et al., 2023). Nanocarriers, with their customizable size, modifiable surface properties, and multifunctionality, provide effective solutions for crossing the BBB and enabling targeted drug delivery (de Almeida Campos et al., 2023). Currently, nanoparticle-based delivery systems serve as a key strategy for crossing the BBB and enabling targeted therapy (Hornok et al., 2022; Song et al., 2024; Zha et al., 2024).

Material selection is crucial in nanocarrier design. Commonly used synthetic materials mainly include lipid-based (such as liposomes, solid lipid nanoparticles), polymer-based (such as PLGA nanoparticles, PCL nanoparticles), inorganic materials (such as gold nanoparticles, silica nanoparticles), and lipid-polymer hybrid nanoparticles (Terstappen et al., 2021). Each material possesses unique characteristics, providing diverse options for developing multifunctional combination therapies.

For small molecule drug delivery, nanocarriers offer significant advantages. Chen et al. (2020) developed a lipoprotein-biomimetic nanocarrier that efficiently delivered cyclosporine A, reducing neuronal injury and inflammation in TBI models with only 1/16 of the free CsA dosage. Sun et al. (2022b) designed red blood cell-coated nanolipid carriers modified with C3 and SS31 peptides to deliver the PARP inhibitor olaparib targeted to neuronal mitochondria, significantly enhancing mitochondrial function and preventing neuronal death. In another study, researchers encapsulated Cl-amidine in self-assembling liposomes modified with reactive oxygen species-responsive polymers and fibrin-binding peptides, enabling targeted delivery to ischemic lesions and stimulus-responsive release (Sun et al., 2023).

In macromolecule delivery, nanocarriers offer unique advantages. Researchers developed polymer nanocomplexes to efficiently deliver siRNA (siRhoA), reducing RhoA protein expression to suppress neuroinflammation and cell apoptosis (Macks et al., 2021). In another study, engineered nanoparticles accumulate in brain tissue at over three times the rate of non-engineered PEG nanoparticles in TBI mouse models (Li et al., 2021b). Brain-derived neurotrophic factor encapsulated in engineered porous silicon nanoparticles significantly reduced lesion volume through the CAQK targeting delivery system (Waggoner et al., 2022).

Smart, responsive drug delivery systems drive innovation in nanocarrier development. Qian et al. developed an injectable curcumin-loaded hydrogel that responds to the TBI microenvironment, releasing drugs to reduce brain edema and inflammation while promoting neuroregeneration through reactive oxygen species scavenging (Qian et al., 2021). Recent research highlights the development of transferrin receptor-targeted minocycline albumin nanoparticles, significantly enhanced drug accumulation in the brain, improving behavioral outcomes, and reducing toxicity (Perumal et al., 2023). A multifunctional nanocomplex called AMEC (curcumin loaded on manganese-doped melanin-like nanoparticles modified with targeting peptides) targets brain injury sites after intravenous injection, effectively treating TBI through antioxidant and anti-inflammatory mechanisms (Sun et al., 2022a).

These findings indicate that nanocarrier synthesis improves the therapeutic effect of a single drug and serves as an important platform for developing multifunctional combinatorial therapies. Rational design of carrier materials and surface modifications enables the synergistic delivery of multiple therapeutics, significantly improving the therapeutic effect of TBI.

Artificial intelligence-based treatment decision support

Artificial intelligence (AI), a rapidly developing field, provides new approaches for TBI diagnosis and treatment through computational methods such as machine learning and deep learning (**Table 2**; Stonko et al., 2021). In complex clinical decisions involving multiple factors and high uncertainty, AI technology demonstrates unique advantages in providing more precise risk assessment and treatment plans (Farzaneh et al., 2021).

AI is crucial in designing TBI combination therapy. Through analyzing large clinical datasets, AI systems predict the interaction between different therapeutic drugs, providing a scientific basis for formulating personalized combination regimens (Segato et al., 2020). For example, upon detecting elevated intracranial pressure, AI recommends appropriate drug combinations and optimizes them through continuous learning and tracking of treatment effects (Romm and Tsigelny, 2020; Zoerle et al., 2024). This dynamic optimization is crucial for complex combination therapies, providing reliable decision support for clinicians.

In neurological intensive care, the data processing capability of AI shows unique advantages. AI systems continuously collect and analyze multidimensional monitoring data, applying deep learning to identify key prognostic indicators and support clinical decision-making (Zhu, 2024). This AI-based comprehensive analysis integrates physiological indicators with multi-source data, including imaging and genomics, to construct a comprehensive evaluation system that guides the formulation and adjustment of combination therapy plans (Pierre et al., 2024). AI technology has achieved automated assessment of clinical scale scores from videos, such as motor function assessment for patients with Parkinson's disease (Lanotte et al., 2023).

AI-driven TBI rehabilitation therapy is evolving into a comprehensive, systematized approach with diverse applications. Integrating computer-assisted training, robot-assisted rehabilitation, virtual reality technology, and smart mobile devices enhanced precision and personalized therapy (Wang et al., 2021). AI systems can adjust training parameters in real-time based on patient recovery progress, optimize rehabilitation plans through intelligent algorithms, and enable remote monitoring to ensure continuous and effective treatment.

In terms of drug therapy, AI technology is transforming traditional medication patterns (Zhang et al., 2024b). Through machine learning algorithms, AI analyzes extensive clinical medication data to predict the efficacy and potential risks of different drug combinations

Table 2 | **Applications of artificial intelligence in traumatic brain injury management**

Clinical domains	AI technologies	Clinical applications	Expected outcomes	Challenges and limitations
Diagnosis and assessment	Deep learning, computer vision, natural language processing	Imaging analysis, clinical data interpretation, severity assessment	Faster diagnosis, improved accuracy, standardized evaluation	Data quality variability, limited validation, integration complexity
Treatment planning	Machine learning, expert systems, predictive analytics	Drug combination optimization, treatment protocol selection, risk stratification	Personalized treatment, reduced complications, better outcomes	Algorithm transparency, clinical acceptance, real-time adaptation
Monitoring and management	Real-time analytics, neural networks, pattern recognition	Vital signs monitoring, complication prediction, treatment response tracking	Early warning, timely intervention, dynamic adjustment	System stability, alert fatigue, data overload
Rehabilitation support	Reinforcement learning, adaptive algorithms, virtual reality	Exercise guidance, progress tracking, cognitive training	Optimized recovery, enhanced engagement, better compliance	Technical accessibility, cost considerations, user adaptation
Outcome prediction	Statistical learning, survival analysis, and multi-modal fusion	Prognosis prediction, recovery trajectory, long-term outcomes	Better planning, resource optimization, family guidance	Model accuracy, prediction reliability, external validity

This table summarizes the multidimensional applications of AI in the management of traumatic brain injury (TBI). It comprehensively examines AI technologies across five clinical domains: diagnosis and assessment, treatment planning, monitoring and management, rehabilitation support, and outcome prediction. The table details specific applications, expected outcomes, and associated challenges and limitations. This systematic review provides a reference framework for healthcare institutions implementing AI-assisted TBI diagnosis and treatment, offering valuable guidance for the standardized application of AI technologies in clinical practice. The detailed analysis supports the integration of AI-based solutions in TBI care while acknowledging current technological constraints and identifying areas for future development. AI: Artificial Intelligence; TBI: traumatic brain injury.

after TBI, providing key insights for doctors to formulate personalized medication plans (Basile et al., 2019; Zhu, 2020). Meanwhile, AI can also screen innovative drug combinations through deep learning models, discovering more effective treatment strategies crucial for managing complex TBI (Jiménez-Luna et al., 2021).

As technology advances, AI plays an increasingly integral role in TBI combination therapy. Multi-modal data fusion enables AI-driven patient evaluations, real-time decision support systems provide timely clinical insights, and telemedicine platforms break through geographical limitations, optimizing the utilization of expert resources (Reddy et al., 2024). These advancements optimize existing treatment strategies and provide possibilities for discovering new treatment combinations. However, the application of AI-assisted decision-making systems requires continuous clinical validation and refinement (Vatansever et al., 2021). Integrating clinical experience with AI recommendations is key to ensuring the scientific validity and safety of treatment plans. As AI technology advances and clinical data accumulates, its role in TBI precision treatment and personalized medicine will continue to expand.

Sequential Combination Strategies

Given the temporal progression of TBI pathophysiology, the importance of stage-specific sequential treatment strategies is highlighted. During the acute phase, when BBB disruption and neuroinflammation dominate, therapeutic approaches prioritize BBB protection and inflammation modulation (Ondruschka et al., 2018). This phase involves the combined administration of BBB stabilizers—such as Claudin-5 agonists and matrix metalloproteinase inhibitors—along with targeted anti-inflammatory agents that regulate microglial activation and cytokine signaling. Additionally, the application of osmotic stabilizers and endothelial protective compounds may contribute to maintaining BBB integrity. As the pathology advances to the subacute phase, characterized by secondary injury cascades and oxidative stress, the therapeutic focus shifts toward neuroprotection and antioxidant interventions. This phase incorporates neurotrophic factors, anti-apoptotic agents, and free radical scavengers while targeting mitochondrial dysfunction and cellular energy deficits (Modi et al., 2024). The chronic phase primarily focuses on neuroregeneration and functional recovery, integrating growth factors and stem cell-based therapies to enhance neuroplasticity and axonal regeneration alongside targeted rehabilitation programs. For example, stem cell factor combined with granulocyte colony-stimulating factor therapy has demonstrated strong efficacy in improving long-term functional outcomes, enhancing neuroplasticity, restoring neural structural network balance disrupted by severe TBI, and promoting myelin regeneration, thereby supporting brain repair during the chronic phase of TBI (Qiu et al., 2020, 2023). This sequential treatment approach is consistent with the dynamic progression of TBI pathophysiology and may provide more effective outcomes than the traditional single-stage treatments.

Future Directions of Combination Therapies

While combination therapy offers unique advantages in managing the dynamic pathological processes after TBI, its clinical application remains challenging. The primary challenge is optimizing the sequence of drug administration (Borgen et al., 2020). For example, when anti-inflammatory drugs are combined with stem cell therapy, administering anti-inflammatory drugs too early may affect stem cell survival and function. Additionally, some nanocarriers may alter the pharmacokinetic properties of co-administered therapies. Therefore, researchers must carefully evaluate the mechanisms of action, therapeutic targets, and time-dependent effects of various treatment modalities.

Drug interactions present a significant challenge in combination therapy. When multiple drugs are administered together, they may exhibit pharmacokinetic and pharmacodynamic interactions (Calzetta et al., 2024). Some drugs can affect the cytochrome P450 enzyme system, thereby modifying the metabolic processing of co-administered agents (O'Meara et al., 2024). Additionally, different drugs may interact at the receptor level, producing competitive or synergistic effects, which may lead to unexpected therapeutic outcomes or adverse reactions. The complexity of safety assessment increases when combination therapy includes emerging treatments such as stem cells, exosomes, and nanomaterials. Stem cell therapy can modulate the immune response of the body, potentially altering the safety profile of co-administered therapeutic approaches. Similarly, the use of nanomaterials may influence drug distribution within tissues, introducing additional safety considerations. Addressing these challenges requires the development of a comprehensive safety assessment framework, including systematic toxicology studies, long-term safety monitoring, immunogenicity assessments, biocompatibility research, and investigations into drug interaction mechanisms (Ahmad et al., 2022; Souto et al., 2024).

The complexity of delivery systems presents another significant challenge in combination therapy (Mashkouri et al., 2016). When multiple therapeutic approaches are combined, selecting the appropriate delivery routes becomes more challenging. Some drugs may be suitable for systemic administration, while others may require localized delivery to achieve optimal effects. A critical issue is coordinating these different delivery routes to ensure that the therapeutic approaches exert synergistic effects at the target sites. In the engineering design of drug delivery systems, several technical challenges must be addressed when using nanocarriers for the co-delivery of multiple drugs. These challenges include optimizing drug loading capacity and encapsulation efficiency, controlling drug release kinetics, matching material degradation properties, designing effective surface modification strategies, and achieving precise targeting functions (Kemp et al., 2016). Additionally, BBB permeability undergoes dynamic changes after injury, necessitating delivery systems that adapt to different pathological states. Furthermore, considering the unique characteristics of brain tissue, these systems must exhibit high biocompatibility and precisely controlled degradation properties.

Despite numerous challenges, combination therapy holds significant promise for advancing TBI treatment. Technological advancements continue to enhance diagnostic capabilities, providing more precise condition assessments. Innovations such as advanced imaging technologies, high-throughput detection platforms, the discovery of novel biomarkers, and the refinement of AI-assisted diagnostic systems will contribute to the development of individualized treatment strategies (Reddy et al., 2024). Significant advancements in drug delivery technology have led to the development of smart, responsive materials, multifunctional nanocarriers, optimized targeting strategies, and innovative delivery devices. The diversification of therapeutic approaches—including the emergence of novel drugs, the maturation of stem cell therapy, the clinical translation of gene therapy, and the exploration of immunotherapy—continues to enhance the potential of combination therapy.

Looking ahead, as the understanding of TBI pathogenesis grows and multidisciplinary integration increases, combination therapy is expected to play an increasingly significant role in TBI treatment. Advancing fundamental research, addressing existing technical barriers, standardizing treatment protocols, and continually optimizing clinical strategies will be essential in maximizing its therapeutic potential. These efforts hold promise for improving patient outcomes, significantly enhancing prognosis, and ultimately elevating the quality of life for individuals with TBI. Furthermore, reforms in the healthcare system will facilitate the adoption and integration of novel treatment modalities, promoting the broader implementation of combination therapy in TBI treatment.

Consideration of Individualized Therapy Based on Imaging and Biomarkers

The development of individualized treatment plans presents substantial challenges. Patients with TBI exhibit significant differences in injury mechanisms, anatomical location and severity, progression of secondary injury, types and severity of complications, and baseline health status (Covington and Duff, 2021). This high degree of heterogeneity requires tailoring combination therapy to individual patient needs. However, several challenges in this area, including the absence of reliable prognostic models, incomplete biomarker frameworks, lack of standardized treatment response metrics, and difficulties in establishing uniform individualized treatment protocols (Jaganathan and Sullivan, 2020). Achieving truly personalized treatment requires the development of more comprehensive patient stratification strategies, identification of novel biomarkers, and integration of artificial intelligence and other advanced technologies to assist in treatment planning (**Table 3**).

Table 3 | Imaging and biomarker-based patient stratification for personalized TBI treatment

Categories	Markers/Techniques	Clinical indicators	Treatment implications	Monitoring parameters
Blood biomarkers	S100B, NSE, GFAP, UCH-L1, NF-L, IL-6, TNF-α	Injury severity, neural damage extent, inflammatory status, disease progression	Neuroprotective strategy selection, anti-inflammatory intervention timing, treatment intensity adjustment	Marker level dynamics, treatment response, complication risks
CSF markers	Tau protein, amyloid-β, NGF, BDNF, microRNA profiles	Neural repair potential, cognitive prognosis, molecular typing	Regenerative therapy planning, cognitive intervention design, treatment duration optimization	CSF protein levels, neural repair indicators, cognitive assessments
Conventional imaging	CT (Marshall classification), MRI (T1/T2/FLAIR), SWI	Lesion morphology, hemorrhage detection, edema assessment	Surgical planning, medical management, rehabilitation timing	Lesion evolution, edema resolution, structural recovery
Advanced imaging	Functional MRI, diffusion tensor imaging, network connectivity analysis	Network dysfunction, white matter integrity, functional connectivity	Neuromodulation strategies, rehabilitation protocols, cognitive training design	Network reorganization, tract integrity, functional recovery
Integrated analysis	Multimodal imaging, combined biomarker panels, AI-based integration	Comprehensive assessment, risk stratification, outcome prediction	Personalized protocol design, treatment modification, long-term planning	Multiple parameters, treatment efficacy, prognostic indicators

This comprehensive table presents a systematic framework for integrating imaging techniques and biomarker assessments in personalized treatment strategies for TBI. It categorizes and analyzes five key domains: blood biomarkers, CSF markers, conventional imaging, advanced imaging, and integrated analysis. For each category, the table details specific markers or techniques, clinical indicators, treatment implications, and monitoring parameters. This structured approach provides clinicians with a practical guide for patient stratification and treatment personalization. The integration of both traditional and advanced diagnostic tools, alongside their clinical applications, offers a robust foundation for evidence-based decision-making in individualized TBI care. Notably, the table emphasizes the importance of dynamic monitoring and treatment adjustments based on multiple parameters, facilitating the optimization of therapeutic strategies throughout the treatment course. This systematic organization of diagnostic and monitoring tools supports the development of more precise and effective personalized treatment protocols, potentially improving patient outcomes through targeted interventions based on specific biomarker and imaging profiles. AI: Artificial intelligence; BDNF: brain-derived neurotrophic factor; CSF: cerebrospinal fluid; CT: computed tomography; FLAIR: fluid-attenuated inversion recovery; GFAP: glial fibrillary acidic protein; IL-6: interleukin-6; MRI: magnetic resonance imaging; NF-L: neurofilament light chain; NGF: nerve growth factor; NSE: neuron-specific enolase; SWI: susceptibility-weighted imaging; TBI: traumatic brain injury; TNF-α: tumor necrosis factor-alpha; UCH-L1: ubiquitin C-terminal hydrolase L1.

Patient heterogeneity in traumatic brain injury

Homogeneity in TBI remains a significant challenge to translating research into effective therapies, as it involves extensive variability in single-cell responses to both external (e.g., TBI models) and internal factors (e.g., sex, time, and brain regions) (Jha et al., 2024). From a mechanistic perspective, patients may sustain different types of injuries, including direct impact, penetrating injury, acceleration-deceleration forces, or blast-related injury, each of which induces unique pathological changes (Dixon, 2017). Direct impacts can lead to focal injuries, such as contusions and intracerebral hematomas, while acceleration-deceleration forces typically result in diffuse axonal injury, compromising white matter connectivity. Blast-related injuries are more complex, potentially causing both primary blast wave-induced damage and secondary tissue injury, each associated with distinct pathological changes (Bandak et al., 2015).

Variability in the anatomical location of injury is equally significant, with damage potentially affecting the cerebral cortex, brainstem, basal ganglia, cerebellum, or multiple regions simultaneously. Injuries in different brain areas lead to distinct functional deficits. For example, frontal lobe damage is often associated with executive dysfunction and personality changes, and temporal lobe injury can affect memory and emotional processing, occipital lobe damage typically leads to visual disturbances, and brainstem injuries may disrupt the regulation of vital signs (Jones and Graff-Radford, 2021). Diffuse axonal injury is particularly unique for its ability to disrupt information transmission across multiple brain regions, often leading to widespread cognitive dysfunction (Armstrong et al., 2024; Zhang et al., 2024a).

TBI severity is traditionally classified as mild, moderate, or severe based on the GCS. However, patients with identical GCS scores can present with significantly different clinical symptoms and prognoses (Tenovuo et al., 2021). For example, while some patients with mild TBI recover within weeks, others develop persistent post-traumatic syndrome characterized by chronic headaches, dizziness, cognitive impairments, and emotional disturbances (Jenkins, 2023).

The heterogeneity of clinical presentation in TBI is particularly complex. During the acute phase, symptoms may include altered consciousness levels, neurological deficits, and autonomic dysfunction; however, their severity and combination vary among individuals (Ahmed, 2022). Some patients primarily experience motor function impairments while maintaining relatively preserved cognitive function, while others exhibit significant cognitive decline despite relatively intact motor function. Emotional and behavioral changes following TBI vary significantly among individuals, ranging from mild emotional instability to severe personality changes (Howlett et al., 2022). Extensive research supports this heterogeneity. A study involving 1057 patients with TBI found that, while many demonstrated normal cognitive function 6 months post-injury, those with cognitive impairments exhibited diverse patterns of dysfunction (Bryant et al., 2023). These differences are associated with multiple factors, including age, sex, genetic background, medical history, and pre-injury cognitive reserve (Giordano et al., 2020).

Furthermore, the progression of secondary injury varies significantly among individuals. Some patients experience severe intracranial pressure elevation and cerebral edema, while others remain relatively stable. Variability in inflammatory response intensity, metabolic changes, and neuroplasticity potential collectively influence the recovery trajectory. A previous study suggests that microbiome dysregulation and differences in peripheral immune response may contribute to TBI progression, further increasing patient heterogeneity (Ullah et al., 2024).

The significant heterogeneity of TBI presents important challenges in clinical practice. Traditional "one-size-fits-all" treatment approaches often fail to address the adverse needs of patients, which may explain the suboptimal outcomes observed in many TBI clinical trials. Recognizing and accounting for this heterogeneity is crucial for advancing personalized treatment strategies and optimizing therapeutic outcomes.

Biomarker-based and imaging-based stratification

Given the significant heterogeneity among patients with TBI, developing precise patient stratification strategies has become increasingly essential. Currently, stratification approaches based on biomarkers and imaging features are undergoing continuous refinement. In biomarker research, blood-based markers have gained attention due to their ease of acquisition (Hayes-Larson et al., 2024). The S100B protein, a specific marker of astrocyte injury, demonstrates a strong correlation between serum levels and injury severity. Studies indicate that peak S100B levels within 6 hours post-injury can serve as a prognostic indicator (Jenkins et al., 2023). Similarly, neuron-specific enolase reflects neuronal injury extent, with sustained elevated levels often associated with poor prognosis (Freitas et al., 2024). GFAP, a specific marker of astrocytes, reflects injury severity and helps differentiate intracranial hemorrhage from ischemic injury (Abdelhak et al., 2022).

Recent studies have demonstrated the discovery and validation of novel blood biomarkers. Neurofilament light chain protein, an axonal injury marker, has demonstrated significant value in diagnosing and monitoring diffuse axonal injury (Hossain et al., 2023). Ubiquitin C-terminal hydrolase L1, a specific marker of neuronal injury, has been approved by the FDA for use in combination with GFAP to evaluate the necessity of CT scans (Oris et al., 2024). Additionally, dynamic changes in inflammatory markers such as IL-6 and TNF-α can indicate the progression of secondary injury (Kazakova et al., 2021).

While cerebrospinal fluid (CSF) biomarkers are more difficult to obtain, they offer direct insights into central nervous system injury. Changes in CSF tau protein and amyloid-β levels can help predict cognitive function outcomes (Rubenstein et al., 2024). The detection of nerve growth

factor and brain-derived neurotrophic factor aids in assessing neural repair potential (Sims et al., 2022). Emerging research suggests that shifts in CSF microRNA profiles may provide more precise patient stratification (Lusardi et al., 2021; Dahal et al., 2024).

For imaging-based classification, traditional CT and MRI provide fundamental morphological categorization. The Marshall CT classification remains widely used in clinical practice, stratifying patients into diffuse injury types I–IV, along with surgical and non-surgical lesions (Asim et al., 2021). The Rotterdam scoring system integrates factors such as basal cistern compression and midline shift, providing a more precise prognostic assessment (Elkbuli et al., 2021). Conventional MRI sequences, such as T1, T2, and FLAIR, provide detailed anatomical insights into injury characteristics, forming a foundation for precise classification (Lee, 2020). Susceptibility-weighted imaging offers unique advantages in detecting microhemorrhages, enhancing the assessment of diffuse axonal injury severity (Jaafari et al., 2024).

Functional magnetic resonance imaging (fMRI) has provided novel insights into brain network alterations. Resting-state fMRI studies have revealed distinct patterns of functional connectivity changes across different TBI subtypes (Palacios et al., 2013; Wei et al., 2024). For example, disruptions in default mode network connectivity are associated with cognitive function prognosis, while motor network alterations correlate with motor function recovery. Diffusion tensor imaging (DTI) facilitates the quantitative assessment of axonal injury by evaluating white matter tract integrity (Wang et al., 2024a). Changes in fractional anisotropy (FA) values indicate injury severity and serve as predictors of long-term outcomes.

The integration of multimodal imaging analysis represents the future direction of classification methods. Machine learning algorithms can combine data from CT, conventional MRI, functional imaging, and DTI to create more accurate predictive models (Zhang et al., 2020). For instance, studies employing deep learning techniques to analyze multimodal imaging features have successfully categorized TBI patients into distinct prognostic subgroups (Lin et al., 2023a; Hibi et al., 2024). This integrated analysis not only enhances classification accuracy but also aids in the identification of new imaging biomarkers.

Recent research has begun to integrate biomarker and imaging features for analysis. For example, combining serum GFAP levels with DTI parameters can more accurately predict cognitive function prognosis (Nasrallah et al., 2023). Correlating dynamic changes in multiple biomarkers with alterations in functional connectivity helps improve our understanding of injury mechanisms and recovery processes. This multidimensional stratification strategy not only enhances classification accuracy but also provides a more reliable foundation for developing personalized treatment plans.

Based on the findings from these stratified approaches, clinicians can more accurately assess patient status, predict disease progression, and tailor personalized treatment options. However, these methods still require validation in larger prospective studies, particularly across different populations and clinical settings. Additionally, cost-effectiveness analysis is an important factor that must be considered to promote the adoption of these stratification methods.

Precision stratified treatment

Precise stratification based on biomarkers and imaging features provides essential guidance for personalized treatment of TBI. This stratification strategy allows for the development of targeted treatment plans tailored to patients' specific characteristics. For patients primarily exhibiting axonal injury, the treatment focus should be on neuroprotection and axonal regeneration. Conversely, for those with primarily neuronal injury, indicated by elevated levels of serum neuron-specific enolase and UCH-L1, treatment strategies should emphasize neuronal protection (Silvestro et al., 2024). These patients may also be more responsive to hypothermia treatment, as research has shown that precise control of target temperatures can significantly improve outcomes. For instance, one study demonstrated that temperature-sensitive mesenchymal stem cells combined with mild hypothermia could reduce neurological deficits in rats with severe TBI (Song et al., 2020). Additionally, patient groups exhibiting significant inflammatory responses, as evidenced by elevated levels of inflammatory factors such as IL-6 and TNF-α, may require more aggressive anti-inflammatory treatments (Kalra et al., 2022). Clinical practice has indicated that the early use of selective anti-inflammatory drugs in these patients may help reduce secondary injury.

Patients exhibiting impairment in the default mode network on functional imaging may encounter greater challenges in cognitive rehabilitation. For these individuals, early initiation of cognitive training combined with neuromodulation techniques, such as transcranial magnetic stimulation, may lead to improved rehabilitation outcomes (Yen et al., 2024). Additionally, transcranial electrical stimulation (tES) can be integrated with real-time fMRI to create a closed-loop tES-fMRI system for personalized neuromodulation (Soleimani et al., 2023). The combination of transcranial direct current stimulation with computerized cognitive training has also demonstrated significant improvements in cognitive flexibility tests for TBI patients (Afsharian et al., 2024).

For patients with primarily focal injuries identified through imaging, the timing of surgery and the selection of surgical approaches must be more individualized (Marehbian et al., 2017). By integrating CT features, biomarker levels, and clinical presentations, clinicians can develop tailored surgical decision-making plans that significantly enhance surgical outcomes.

Stratification-based treatment adjustment strategies also involve personalized medication selection. For example, patients with persistently elevated S100B levels may require more aggressive neuroprotective treatments, while those with abnormalities in serum osmolarity may need more precise fluid management strategies (Menon and Ercole, 2017). Patients whose treatment plans were adjusted based on dynamic biomarker changes showed significantly better outcomes compared to those who received standardized treatment.

Personalizing rehabilitation strategies is equally crucial. Patients showing motor pathway damage on DTI may require earlier and more intensive physical therapy, while those with abnormalities in cognitive network functional connectivity may need additional cognitive rehabilitation training. Individualizing treatment intensity is also essential. Patients whose biomarkers indicate severe and progressive injury may necessitate more intensive monitoring and aggressive interventions.

Long-term follow-up strategies should also be tailored based on stratification results (Chard et al., 2021). Different patient subtypes carry varying risks of long-term complications, making it necessary to develop differentiated follow-up plans. In clinical practice, implementing personalized treatment strategies requires close coordination among multidisciplinary teams. For instance, the neurosurgery, neurology, and rehabilitation departments should collaborate to create comprehensive treatment plans based on patient stratification results.

Limitations

Insufficient research in chronic phase of traumatic brain injury

Research into the chronic phase following TBI represents a critical yet understudied area in neuroscience and clinical medicine. While the medical community has focused extensively on the acute and subacute phases of TBI in recent years, systematic investigations into the long-term evolution of neurological function and the underlying pathological processes remain notably limited (Wilson et al., 2017). The shortcomings in this research area primarily manifest as insufficient mechanistic studies on neurodegenerative changes in the chronic phase, a lack of long-term follow-up data, and an absence of targeted intervention strategies (Yiannopoulou et al., 2019). Future research must establish long-term follow-up cohorts and employ a comprehensive, multidisciplinary approach, integrating neuroimaging, molecular biology, and neurophysiology to systematically track patients' neurological functional trajectories. This approach should aim to analyze key pathological processes such as neuroinflammation, synaptic plasticity reconstruction, and neuronal apoptosis, ultimately leading to more targeted and individualized intervention protocols for chronic TBI patients (Inoue et al., 2005). Moreover, the psychological changes that patients experience constitute an equally important dimension of research. Issues related to psychosocial adaptation following TBI—such as cognitive dysfunction, emotional regulation disorders, and personality changes—require increased attention and systematic investigation (Stocchetti and Zanier, 2016; Doser et al., 2018). This represents not only a significant scientific research proposition but also a considerable clinical challenge that necessitates sustained attention and in-depth exploration across multiple disciplines.

Limitations in stem cell applications

Stem cell therapy has certain limitations (**Figure 5**). Key challenges in stem cell applications include tumorigenicity, immunogenicity, and heterogeneity (Zhong et al., 2022). Stem cell heterogeneity manifests in various functional modes, such as proliferation capacity, transdifferentiation, immunophenotype, and paracrine and microvesicle mechanisms involving secretome-derived products. Notably, differences in these mechanisms exist between stem cells derived from different sources (Costa et al., 2021). Immune rejection is another critical issue in cell therapy, with the immunogenicity of autologous iPSCs remaining a topic of debate (Sharkis et al., 2012). The tumorigenic potential of stem cell therapy also warrants careful consideration, as the use of stem cells in treatment is still regarded as a high-risk therapeutic approach due to the possibility of tumor induction following cellular transplantation (Aly, 2020). Pluripotent stem cells, including iPSCs, share conserved gene expression networks with cancer cells. During the preparation of transplanted cells, the presence of incompletely reprogrammed cells, undifferentiated iPSCs, and even differentiated iPSCs can heighten the carcinogenic potential of therapeutic cells (Zhong et al., 2022). Several direct and indirect mechanisms of mesenchymal stem cell (MSC) action may contribute to cellular cancer origins, including the induction of epithelial-mesenchymal transition (EMT), which can generate cancer stem cell-like states in cancer cells, potentially serving as tumorigenic progenitors in sarcomas and maintaining cancer stemness (Li et al., 2021a).

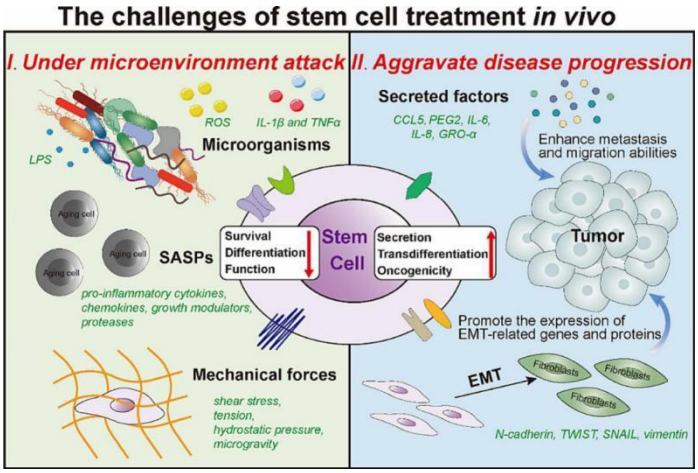

Figure 5 | **Challenges of *in vivo* stem cell treatment.**
This schematic illustrates the dual challenges faced by stem cells during *in vivo* treatment (Han et al., 2024). The left panel depicts microenvironmental adversities from various sources, including microorganisms that release ROS, IL-1β, and TNFα; senescent cells that produce SASP factors containing pro-inflammatory cytokines, chemokines, and proteases; and mechanical forces such as shear stress, tension, and hydrostatic pressure. Collectively, these factors diminish stem cell survival, differentiation, and function. The right panel illustrates how stem cells can potentially exacerbate disease progression through the secretion of factors such as CCL5, PEG2, IL-6, IL-8, and GRO-α, which enhance tumor metastasis and migration. Additionally, these factors promote EMT by regulating gene expression of N-cadherin, TWIST, SNAIL, and vimentin, leading to fibroblast transformation. Together, these challenges underscore the complexities and potential risks associated with *in vivo* stem cell therapeutic applications. CCL5: Chemokine (C–C motif) ligand 5; EMT: epithelial-mesenchymal transition; GRO-α: growth-regulated oncogene alpha; IL-1β: interleukin-1 beta; IL-6: interleukin-6; IL-8: interleukin-8; PEG2: prostaglandin E2; ROS: reactive oxygen species; SASP: senescence-associated secretory phenotype; SNAIL: snail family transcriptional repressor; TNFα: tumor necrosis factor alpha; TWIST: twist family bHLH transcription factor. © 2023 Elsevier B.V. All rights reserved (Han et al., 2024; License Number: 5924810208483).

Challenges in technology integration

Technology integration is a crucial component of combined treatment for TBI, yet it continues to face numerous technical challenges. The most pressing issue is the need for synergistic coordination among various therapeutic approaches, which involves precise timing arrangements and optimization of dosage across different modalities. For instance, when attempting to combine pharmacological treatments with physical therapies—such as hyperbaric oxygen, focused ultrasound, and hypothermia therapy—clinicians often struggle to determine the optimal therapeutic windows and intervention sequences (Krogager Mathiasen et al., 2020). This temporal uncertainty can lead to significant variations in treatment efficacy and even unexpected antagonistic effects. In particular, during the acute phase of treatment, some medications may affect the efficacy of physical therapy, while physical treatments might alter the distribution and metabolic characteristics of drugs. Furthermore, optimizing dosages in multimodal therapy presents significant challenges (Cruz Navarro et al., 2022). For example, coordinating the pressure and duration of hyperbaric oxygen treatment with drug administration protocols, or aligning focused ultrasound energy parameters with drug delivery system characteristics, remains under-researched. In terms of therapeutic monitoring, the lack of unified evaluation standards and reliable real-time monitoring methods complicates the accurate assessment of the synergistic effects of different treatment approaches. These monitoring limitations directly impact the ability to make timely adjustments and optimize treatment protocols. Additionally, there are no established standardized implementation procedures or quality control systems for multimodal therapy, leading to significant variations in treatment protocols across different medical centers. This variability not only affects the reproducibility and reliability of treatment outcomes but also poses challenges for multi-center clinical studies. Compatibility issues between different treatment equipment and technical platforms also require resolution, including the integration of data from monitoring devices and the real-time adjustment of treatment parameters. Addressing these issues necessitates the development of unified technical standards and operational protocols. Overall, these technical integration challenges not only hinder the clinical application of existing combined treatment protocols but also constrain the development and optimization of novel therapeutic strategies.

Limited research on patients with comorbidities

Patients with comorbidities represent a unique and complex population in the treatment of TBI, yet current research on combination therapy strategies for these individuals remains severely insufficient. These patients may simultaneously present with underlying conditions such as diabetes, hypertension, heart disease, coagulation disorders, and hepatorenal dysfunction, all of which significantly impact the development and implementation of treatment plans (Renu et al., 2020; Wang et al., 2024b). For instance, diabetic patients may exhibit significantly different BBB permeability and neuroinflammatory response patterns compared to non-diabetic patients, affecting not only drug distribution in the central nervous system but also potentially altering therapeutic responses (Sheikh et al., 2022). In patients with coagulation disorders, the safety and efficacy of certain therapeutic approaches, such as hyperbaric oxygen and stem cell therapy, remain insufficiently validated. Those with hepatorenal dysfunction display unique characteristics in drug metabolism and clearance, which directly influence the design of combination drug regimens (Lai et al., 2022b). Moreover, elderly patients with comorbidities demonstrate different immune function statuses, tissue repair capacities, and drug tolerances compared to individuals with isolated brain injuries. However, systematic studies targeting these special populations are currently lacking (Porceddu and Haddad, 2017; Taylor et al., 2023). Many clinical trials often exclude patients with comorbidities as a criterion, resulting in a body of research that inadequately informs the treatment of these complex cases. This limitation not only hinders the individualization of treatment plans but also increases therapeutic risks in clinical practice. Therefore, there is an urgent need for specialized research focused on different types of patients with comorbidities, as well as the establishment of specific therapeutic strategies and risk assessment systems tailored to this population.

Challenge of high costs

While combination therapy strategies for TBI offer significant clinical advantages, their high costs represent a key barrier to clinical implementation. This financial pressure manifests at multiple levels: First, there are substantial equipment and material costs. Professional medical devices, such as hyperbaric oxygen chambers and focused ultrasound systems, require significant investments for both acquisition and maintenance. Additionally, biological products such as exosomes and stem cells come with notably high preparation

costs. Second, human resource expenses are considerable. Multimodal therapy necessitates coordinated efforts from professional medical teams, which include not only neurosurgeons but also specialized therapists, nursing staff, and technical support personnel. This coordination significantly increases operational costs for healthcare facilities (Humphreys et al., 2013). Furthermore, combination therapy often involves longer treatment cycles and more intensive monitoring, which can further elevate the financial burden on patients. Notably, many innovative therapeutic approaches, such as nanodrug delivery systems and AI-assisted diagnostic tools, are challenging to include in medical insurance coverage due to the lack of long-term cost-effectiveness data. As a result, patients may face higher out-of-pocket expenses. In developing countries or regions with limited medical resources, these high treatment costs may prevent many patients from accessing optimal treatment protocols (Adegboyega et al., 2021). Healthcare institutions also encounter uncertainties regarding the return on investment when introducing new technologies, which may dampen their enthusiasm for adopting innovative treatment approaches. More importantly, the current absence of standardized cost-effectiveness evaluation systems complicates the accurate assessment of the comprehensive economic benefits of combination therapy—such as improving patient outcomes, reducing long-term complications, and enhancing quality of life. This lack of assessment not only impacts the formulation of medical insurance policies but also challenges the rational allocation of medical resources. Therefore, establishing scientific cost-effectiveness evaluation systems, optimizing the economic efficiency of treatment protocols, and exploring innovative medical security mechanisms will be crucial for promoting the widespread application of combination therapy strategies.

Conclusion and Outlook

In this review, we summarize existing studies that focus on the bidirectional relationship between BBB disruption and neuroinflammation in TBI, as well as emerging combination therapeutic strategies. Most research indicates that TBI patients exhibit distinct temporal patterns of BBB dysfunction and inflammatory responses across acute, subacute, and chronic phases. The complex interaction between BBB damage and neuroinflammation plays a critical role in the occurrence and progression of TBI through mechanisms involving alterations in energy metabolism, extracellular matrix remodeling, and the crosstalk between bone marrow and central nervous system responses, all of which collectively influence neural repair processes. Recent studies have particularly highlighted the role of calvarial bone marrow in the contribution of immune cells and emphasized the importance of the glycocalyx layer in maintaining BBB integrity. These findings provide new perspectives for therapeutic targeting in TBI treatment.

Among the various therapeutic approaches, combination strategies show particular promise in addressing the multiple pathological processes associated with TBI. Conventional drug combinations, along with emerging technologies such as stem cell therapy, focused ultrasound, and AI-assisted treatment optimization, have demonstrated significant potential for improving treatment outcomes. In contrast, single-target interventions have proven to be of limited efficacy in addressing the complex pathophysiology of TBI. Therefore, adopting integrated therapeutic strategies represents a viable approach for TBI treatment. Novel delivery systems, biomarker-based patient stratification, and personalized medicine are promising developments that facilitate targeted interventions while catering to individual patient needs. However, existing research has primarily focused on the acute and subacute phases of TBI, leaving the mechanisms involved in the chronic phase largely unexplored. Additionally, data on long-term outcomes are lacking. The integration of multiple therapeutic modalities necessitates careful consideration of factors such as timing, dosing, and potential interactions, especially in patients with comorbidities. Consequently, further investigations are required to determine the feasibility and effectiveness of these innovative methods.

Currently, direct studies focusing on the mechanisms of the chronic phase of TBI are still relatively scarce. Additionally, standalone investigations of single therapeutic approaches are insufficient to fully address the complex pathophysiology of TBI. However, AI-based systems show promise in optimizing combination therapy protocols and predicting patient outcomes. Integrating multiple therapeutic modalities through standardized protocols, while also considering cost-effectiveness, will be crucial for advancing clinical applications. The development of more sophisticated monitoring systems and biomarker panels may help optimize treatment timing and selection. Furthermore, the emergence of novel delivery systems, such as engineered exosomes and smart nanocarriers, presents new opportunities for targeted interventions. The potential of these approaches to modulate both BBB integrity and neuroinflammatory responses warrants further investigation.

Looking ahead, several key challenges must be addressed. The standardization of combination therapy protocols, particularly regarding timing and dosing, remains a significant hurdle. Additionally, the high costs associated with advanced therapeutic approaches limit their widespread implementation, highlighting the need for more cost-effective strategies. Patient heterogeneity and the influence of comorbidities on treatment responses necessitate the development of sophisticated stratification approaches. Furthermore, creating better assessment tools for monitoring treatment efficacy and long-term outcomes is crucial. Therefore, further studies are needed to clarify the significance of timing and sequencing in combination therapy, as well as to understand the specific mechanisms involved in chronic phase pathology. We anticipate the emergence of more sophisticated therapeutic strategies that can safely and effectively improve long-term outcomes for TBI patients, while also emphasizing the importance of personalized approaches that account for individual patient characteristics and needs. The continued advancement of our understanding of TBI pathophysiology, combined with technological innovations in therapeutic delivery and monitoring, holds great promise for developing more effective treatment strategies for this challenging condition.

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